6 Introduction of Transmission Lines

It is a two conducting system that provides a medium for the transmission of energy or power. Transmission lines are commonly used in power distribution (at low frequencies) and in communication (at high frequencies). A transmission line consists of two or more parallel conductors used to connect a source to a load. The source may be a generator, transmitter, antenna, oscilloscope, etc. Typical transmission lines include two-wire lines, co-axial cable, parallel plate, and microstrip.

Transmission line problems are solved using electric circuit theory and electromagnetic theory, but in this chapter, we use electric circuit theory because it is easier to deal with mathematically.

In this chapter, we will discuss the derivation of transmission-line equations and characteristics quantities, standing wave ratio, reflection coefficient, the use of the smith chart, various applications of transmission lines, and transients on transmission lines.

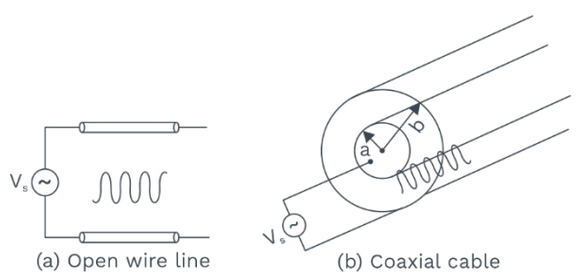
Grey Matter Alert!

A transmission line is a connector that transmits an electromagnetic energy from one point to another point with minimum losses over a wideband frequency.

TYPES OF TRANSMISSION LINES

There are various types of transmission lines:

1) Open-wire line
2) Cables
3) Co-axial line
4) Waveguides



(a) Open wire line (b) Coaxial cable

Fig. 6.1 Types of the Transmission Line

TRANSMISSION LINE PARAMETERS

There are various types of electric parameters associated with the transmission lines.

1) **Resistance:** The resistance is uniformly distributed along the length of the transmission line and its total value depends on the overall length and cross-sectional area of conductors of the transmission lines. It is denoted by R and measured in ohms per unit length.
2) **Inductance:** When the conductor carries the current, the magnetic flux is produced around the conductors. The flux linkages per ampere of current give rise to the effect called the inductance of the transmission line. Inductance is also distributed along the length of the transmission line.
   It is denoted by L and measured in Henry per unit length of the transmission line.
3) **Capacitance:** The transmission line consists of two parallel conductors, separated by a dielectric like air. Due to this, there exists a capacitance associated with the transmission line. Capacitance is also distributed along the length of the transmission line.
   It is denoted by C and measured in Farads per unit length of the transmission line.
4) **Conductance:** If the dielectric between the conductors is not perfect. Hence, a small amount of current flows through the dielectric called displacement current this is nothing but leakage current and gives rise to a leakage conductance associated with the transmission line. It exists between the conductors and are distributed along the length of the transmission line.
   It is denoted by G and measured in mho per unit length of the transmission line.



Rack Your Brain

What are the important parameters of a transmission line?

EQUIVALENT CIRCUIT OF TRANSMISSION LINE

The equivalent circuit of the transmission line is shown below.

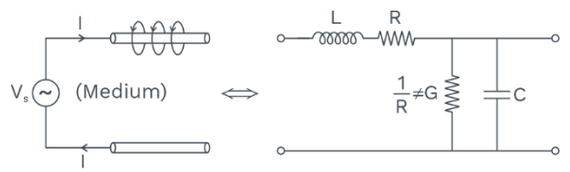


Fig. 6.2 Equivalent Circuit of the Transmission Lines

R = Resistance per unit length (Ω/m)
L = Inductance per unit length (Henry/m)
G = Conductance per unit length (mho/m or S/m)
C = Capacitance per unit length (F/m)
R, L, G and C are primary constants of transmission line.
Series impedance (Z) = R + jωL
Shunt admittance (Y) = G + jωC

Transmission line equation:

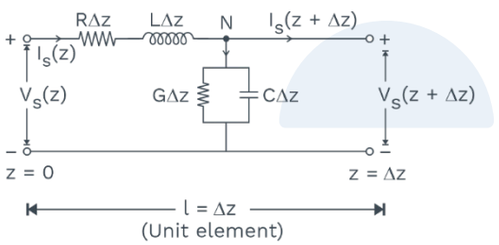


Fig. 6.3 Equivalent Circuit Model of Differential Length Δz of a Two-Conductor Transmission Line

Applying KVL in the outer loop, we get:

$-V_s(z)+I_s(z)\ R\Delta z+j\omega L\ I_s(z)\ \Delta z+V_s(z+\Delta z) = 0$

$V_s(z+\Delta z)-V_s(z) = -I_s(z)(R+j\omega L)\Delta z$

$$\frac{V_s(z+\Delta z)-V_s(z)}{\Delta z} = -I_s(z)(R+j\omega L)$$

$$\lim_{\Delta z \to 0} \frac{V_s(z+\Delta z)-V_s(z)}{\Delta z} = -I_s(z)(R+j\omega L)$$

$$\frac{dV_s(z)}{dz} = -I_s(z)(R+j\omega L) \quad \ldots(i)$$

$$\frac{d^2V_s(z)}{dz^2} = -\frac{dI_s(z)}{dz}(R+j\omega L) \quad \ldots(ii)$$

Rack Your Brain

What do you understand by attenuation constant and phase constant?

Now, applying KCL at node N, we get:

$-I_s(z)+I_s(z+\Delta z)+V_s(z+\Delta z)G\Delta z+V_s(z+\Delta z)j\omega C\Delta z=0$

$I_s(z+\Delta z)-I_s(z)=-V_s(z+\Delta z)\Delta z\,[G+j\omega C]$

or, $$\frac{I_s(z+\Delta z)-I_s(z)}{\Delta z}=-V_s(z+\Delta z)(G+j\omega C)$$

$$\lim_{\Delta z\to 0}\frac{I(z+\Delta z)-I_s(z)}{\Delta z}=\lim_{\Delta z\to 0}\left[-V_s(z+\Delta z)(G+j\omega C)\right]$$

$$\frac{dI_s(z)}{dz}=-V_s(z)(G+j\omega C) \qquad ...(iii)$$

$$\frac{d^2I_s(z)}{dz^2}=-\frac{dV_s(z)}{dz}(G+j\omega C) \qquad ...(iv)$$

From equations iii) and ii), we get:

$$\frac{d^2V_s(z)}{dz^2}=V_s(z)(G+j\omega C)(R+j\omega L)$$

or, $$\frac{d^2V_s(z)}{dz^2}-(R+j\omega L)(G+j\omega C)V_s(z)=0$$

$$\frac{d^2V_s(z)}{dz^2}-\gamma^2V_s(z)=0 \qquad ...(v)$$

This equation is called the transmission line equation (for voltage wave).
From the equations i) and iv), we get:

$$\frac{d^2I_s(z)}{dz^2}=(R+j\omega L)(G+j\omega C)I_s(z)$$

$$\frac{d^2I_s(z)}{dz^2}-\gamma^2I_s(z)=0 \qquad ...(vi)$$

This equation is called the transmission line equation, (for current wave).
Where, γ = propagation constant [m^{-1}]

$$\gamma=\alpha+j\beta=\sqrt{(R+j\omega L)(G+j\omega C)}$$

α = Attenuation constant

β = Phase constant

From equation v), we get:

$$\frac{d^2V_s(z)}{dz^2}-\gamma^2V_s(z)=0$$

$$(D^2-\gamma^2)V_s(z)=0$$

$$D=\pm\gamma$$

$$V_s(z)=V_0^+e^{-\gamma z}+V_0^-e^{\gamma z} \qquad ...(vii)$$

Where

V_0^+ = Forward travelling voltage wave

V_0^- = Reverse travelling voltage wave

From equation vi), we get:

$$\frac{d^2 I_s(z)}{dz^2} - \gamma^2 I_s(z) = 0$$

$$(D^2 - \gamma^2)\, I_s(z) = 0$$

$$D = \pm \gamma$$

$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \qquad \ldots(viii)$$

where I_0^+ = Forward travelling current wave

I_0^- = Reverse travelling current wave

Rack Your Brain

What is the difference between distributed parameters and lumped parameters?

CHARACTERISTICS IMPEDANCE

It is defined as the ratio of the forward travelling voltage wave to the forward travelling current wave at any point of the transmission line.

$$Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

and $$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Definitions

Characteristics impedance:
It is the ratio of the positively travelling voltage wave to the current wave at any point on the line.

Lossless transmission line:

For lossless transmission line:

Condition: R = 0, G = 0

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\alpha + j\beta = \sqrt{(0 + j\omega L)(0 + j\omega C)}$$

$$\alpha + j\beta = j\omega\sqrt{LC}$$

On comparing both sides, we get:

$$\alpha = 0, \quad \beta = \omega\sqrt{LC} \qquad \ldots(i)$$

We know that,

Phase velocity, $$v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}}$$

$$v_p = \frac{1}{\sqrt{LC}}$$

Characteristics impedance,

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

For lossless transmission line, R = G = 0

$$Z_0 = \sqrt{\frac{L}{C}} \quad \text{...(ii)}$$

Multiply β in equation ii) both sides, we get:

$$\beta Z_0 = \omega\sqrt{LC}\sqrt{\frac{L}{C}}$$

$$L = \frac{\beta Z_0}{\omega} \quad \text{...(iii)}$$

By dividing β on both sides in equation ii), we get:

$$\frac{Z_0}{\beta} = \frac{\sqrt{\frac{L}{C}}}{\omega\sqrt{LC}}$$

$$\frac{Z_0}{\beta} = \frac{1}{\omega C}$$

$$C = \frac{\beta}{Z_0\omega} \quad \text{...(iv)}$$

Distortionless transmission line:

The condition for distortionless transmission lines:

$$\frac{R}{L} = \frac{G}{C} \quad \text{...(i)}$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{RG\left(1 + \frac{j\omega L}{R}\right)\left(1 + \frac{j\omega C}{G}\right)}$$

$$= \sqrt{RG}\sqrt{\left(1 + \frac{j\omega C}{G}\right)\left(1 + \frac{j\omega C}{G}\right)}$$

$$= \sqrt{RG}\left(1 + \frac{j\omega C}{G}\right)$$

$$= \sqrt{RG} + j\omega\sqrt{RG \times \frac{C^2}{G^2}}$$

Definitions

Distortionless line:
A distortionless line is one in which the attenuation constant α is frequency independent while the phase constant β is linearly dependent on frequency.

$$= \sqrt{RG} + j\omega\sqrt{RG \times \frac{C}{G} \times \frac{L}{R}} \qquad \left\{\begin{matrix}\text{For distortionless line} \\ \frac{L}{R} = \frac{C}{G}\end{matrix}\right\}$$

$$= \sqrt{RG} + j\omega\sqrt{LC}$$

$$\alpha = \sqrt{RG}, \quad \beta = \omega\sqrt{LC}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= \sqrt{\frac{R}{G}\left(\frac{1 + j\omega\frac{L}{R}}{1 + j\omega\frac{C}{G}}\right)}$$

$$= \sqrt{\frac{R}{G}\left(\frac{1 + j\omega\frac{L}{R}}{1 + j\omega\frac{L}{R}}\right)}$$

$$Z_0 = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} \qquad \text{...(ii)}$$

Phase velocity, $v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$

$$\beta Z_0 = \omega\sqrt{LC} \times \sqrt{\frac{L}{C}}$$

$$L = \frac{\beta Z_0}{\omega} \qquad \text{...(iii)}$$

$$\frac{\beta}{Z_0} = \omega\sqrt{LC}\sqrt{\frac{C}{L}}$$

$$C = \frac{\beta}{\omega Z_0} \qquad \text{...(iv)}$$

$$\alpha Z_0 = \sqrt{RG} \times \sqrt{\frac{R}{G}}$$

$$R = \alpha Z_0 \qquad \text{...(v)}$$

$$\frac{\alpha}{Z_0} = \sqrt{RG}\sqrt{\frac{G}{R}}$$

$$G = \frac{\alpha}{Z_0} \qquad \text{...(vi)}$$

Note:
Lossless lines are distortionless lines, but the distortionless line cannot be lossless.

Low-loss transmission line:
The condition for a low-loss transmission line is:

$R << j\omega L,\ G << j\omega C$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{j\omega L \cdot j\omega C\left(1 + \frac{R}{j\omega L}\right)\left(1 + \frac{G}{j\omega C}\right)}$$

$$= j\omega\sqrt{LC}\left(1 - \frac{jR}{\omega L}\right)^{\frac{1}{2}}\left(1 - \frac{jG}{\omega C}\right)^{\frac{1}{2}}$$

We know, $(1+x)^n \simeq 1 + nx$, if $x << 1$

$$\gamma = \alpha + j\beta = j\omega\sqrt{LC}\left(1 - \frac{jR}{2\omega L}\right)\left(1 - \frac{jG}{2\omega C}\right)$$

$$= j\omega\sqrt{LC}\left(1 - \frac{jR}{2\omega L} - j\frac{G}{2\omega C} - \frac{RG}{4\omega^2 LC}\right)$$

$\because\ RG <<< \omega^2 LC$

So, $\frac{RG}{\omega^2 LC} << 1$

Therefore, $\frac{RG}{4\omega^2 LC}$ terms neglect:

$$\gamma = \alpha + j\beta = j\omega\sqrt{LC} + \omega\sqrt{LC}\,\frac{R}{2\omega L} + \omega\sqrt{LC}\,\frac{G}{2\omega C}$$

or $\alpha + j\beta = \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}} + j\omega\sqrt{LC}$

By comparing both sides, we get:

$$\alpha = \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}},\quad \beta = \omega\sqrt{LC}$$

We know that,

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{j\omega L}{j\omega C}\left[\frac{1 + \frac{R}{j\omega L}}{1 + \frac{G}{j\omega C}}\right]}$$

Rack Your Brain

What are the conditions of R and G for minimum attenuation?

$$= \sqrt{\frac{L}{C}}\left(1 - j\frac{R}{\omega L}\right)^{\frac{1}{2}}\left[1 - j\frac{G}{\omega C}\right]^{-\frac{1}{2}}$$

$$Z_0 = \sqrt{\frac{L}{C}}\left(1 - j\frac{R}{2\omega L}\right)\left(1 + j\frac{G}{2\omega C}\right)$$

$$Z_0 = \sqrt{\frac{L}{C}}\left[1 - j\frac{R}{2\omega L} + j\frac{G}{2\omega C} + \frac{RG}{4\omega^2 LC}\right] \quad \{RG <<< 4\omega^2 LC\}$$

↑ Neglect

$$Z_0 = \sqrt{\frac{L}{C}} - j\frac{R}{2\omega L}\sqrt{\frac{L}{C}} + j\frac{G}{2\omega C}\sqrt{\frac{L}{C}}$$

For Z_0 to be real, $Img(Z_0) = 0$

$$\frac{-R}{2\omega L}\sqrt{\frac{L}{C}} + \frac{G}{2\omega C}\sqrt{\frac{L}{C}} = 0$$

or, $$\frac{R}{2\omega L}\sqrt{\frac{L}{C}} = \frac{G}{2\omega C}\sqrt{\frac{L}{C}}$$

$$\frac{R}{L} = \frac{G}{C}$$

A low-loss transmission line with real characteristics impedance is a distortionless transmission line.

Grey Matter Alert!

A transmission line is called lossless when:

i) The conductors of the transmission lines are perfect.

ii) The conductors separated by the dielectric medium is lossless.
i.e. : R = G = 0

Example 1: An airline has a characteristic impedance of 70Ω and a phase constant of 3 rad/m at 100 MHz. Calculate the inductance per meter and the capacitance per meter of the line.

Sol: Given, $Z_0 = 70\ \Omega$

$\beta = 3$ rad/m

f = 100 MHz

Airline means air is filled between two conductors,

Lossless line: R = G = 0

$$Z_0 = 70 = \sqrt{\frac{L}{C}} \qquad ...(i)$$

$$\beta = 3 = \omega\sqrt{LC} \qquad ...(ii)$$

By multiplying equations i) and equation ii), we get:

$$70 \times 3 = \sqrt{\frac{L}{C}} \times \omega\sqrt{LC}$$

$$210 = \omega L$$

$$L = \frac{210}{\omega} = \frac{210}{2\pi \times 100 \times 10^6} = 33.43 \times 10^{-8} \text{ H / m}$$

or L = 334.3 nH/m

Dividing equation i) by equation ii), we get:

$$\frac{70}{3} = \sqrt{\frac{L}{C}} \cdot \frac{1}{\omega\sqrt{LC}}$$

$$\frac{70}{3} \times \omega = \frac{1}{C}$$

or, $$C = \frac{3}{70 \times \omega} = \frac{3}{70 \times 2\pi \times 100 \times 10^6}$$

$C = 6.82 \times 10^{-11}$ F/m

C = 68.2 pF/m

Example 2: A transmission line operating at 500 MHz has Z_0 = 80 Ω, α = 0.4 Np/m, β = 1.5 rad/m. Find the line parameters R, L, G and C.

Sol:

Given, Z_0 = 80 Ω

α = 0.04 Np/m

β = 1.5 rad/m

ω = 500 MHz

Here, Z_0 is real and $\alpha \neq 0$, Hence, the transmission line is the distortionless line.

$$\alpha = 0.04 = \sqrt{RG} \quad \text{...(i)}$$

$$\beta = 1.5 = \omega\sqrt{LC} \quad \text{...(ii)}$$

$$Z_0 = 80 = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} \quad \text{...(iii)}$$

Multiply equation i) with equation iii), we get:

$\alpha Z_0 = R$

R = 0.04 × 80 = 3.2 Ω/m

Dividing equation i) by the equation iii), we get:

$$\frac{\alpha}{Z_0} = G$$

$$G = \frac{0.04}{80} = 5 \times 10^{-4} \text{ ℧ / m}$$

Multiplying equation ii) with equation iii), we get:

$\beta Z_0 = \omega L$

$$L = \frac{\beta Z_0}{\omega} = \frac{1.5 \times 30}{2\pi \times 500 \times 10^6} = 38.1 \text{ nH / m}$$

Previous Years' Question

A transmission line whose characteristic impedance is a pure resistance:

a) Must be a lossless line.

b) Must be a distortion-less line.

c) May not be a lossless line.

d) May not be a distortion-less line.

Dividing equation ii) by equation iii), we get:

$$\frac{\beta}{Z_0} = \omega C$$

or, $$C = \frac{\beta}{\omega Z_0} = \frac{1.5}{2\pi \times 500 \times 10^6 \times 80}$$

$$= 5.97 \text{ pF/m}$$

Example 3: A distortionless line has $Z_0 = 60\Omega$, $\alpha = 20$ mNp/m, $v_p = 0.6c$, where c is the speed of light in a vacuum. Find R, L, G and C at 50 MHz.

Sol:

Given,

$Z_0 = 60\Omega$, $\alpha = 20$ mNp/m, $v_p = 0.6c$

$f = 50$ MHz

$$\alpha = 20 \times 10^{-3} = \sqrt{RG} \quad \text{...(i)}$$

$$v_p = 0.6 \times 3 \times 10^8 = \frac{1}{\sqrt{LC}} \quad \text{...(ii)}$$

$$Z_0 = 60 = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} \quad \text{...(iii)}$$

Dividing equation i) by equation iii), we get:

$$\frac{\alpha}{Z_0} = G$$

or $$G = \frac{\alpha}{Z_0} = \frac{20 \times 10^{-3}}{60} = 0.33 \times 10^{-3} \text{ ℧ / m}$$

Multiplying equation i) with equation iii), we get:

$$\alpha Z_0 = R$$

or $$R = \alpha Z_0 = 20 \times 10^{-3} \times 60 = 1.2\,\Omega / \text{m}$$

Multiplying equation ii) with equation iii), we get:

$$v_p Z_0 = \frac{1}{C}$$

or $$C = \frac{1}{v_p Z_0} = \frac{1}{0.6 \times 3 \times 10^8 \times 60} = 92.59\,\text{pF/m}$$

Dividing equation ii) by equation iii), we get:

$$\frac{v_p}{Z_0} = \frac{1}{L}$$

or $$L = \frac{Z_0}{v_p} = \frac{60}{0.6 \times 3 \times 10^8} = 0.33\,\mu\text{H/m}$$

Previous Years' Question

Two co-axial cables 1 and 2 are filled with different dielectric constants ε_{r1} and ε_{r2}, respectively. The ratio of the wavelengths in the two cables $\left(\frac{\lambda_1}{\lambda_2}\right)$ is:

a) $\sqrt{\varepsilon_{r_1} / \varepsilon_{r_2}}$

b) $\sqrt{\varepsilon_{r_2} / \varepsilon_{r_1}}$

c) $\varepsilon_{r_1} / \varepsilon_{r_2}$

d) $\varepsilon_{r_2} / \varepsilon_{r_1}$

Example 4: A telephone line has R = 30Ω/km, L = 100 mH/m, G = 0 and C = 20 μF/km and f = 1 kHz, obtain:

i) The characteristics impedance of line.

ii) The propagation constant.

iii) The phase velocity.

Sol:

i) Given,

R = 30 Ω/km

L = 100 mH/km

G = 0

C = 20 μF/km

f = 1 kHz

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{100 \times 10^{-3}}{20 \times 10^{-6}}} = 70.7\,\Omega$$

ii) Assume the line is lossless.

$$\alpha = \sqrt{RG} = \sqrt{30 \times 0} = 0$$

$$\beta = \omega\sqrt{LC} = 2\pi 10^3 \sqrt{100 \times 10^{-3} \times 20 \times 10^{-6}}$$

$$= 8.88 \text{ rad/km}$$

iii) Phase velocity:

$$v_p = \frac{\omega}{\beta} = \frac{2\pi \times 10^3}{8.88} = 707.20\,\text{km/s}$$

$$\gamma = \alpha + j\beta = 0 + j8.88 = j8.88\,\text{Neper/km}$$

Example 5: A telephone line has the following parameters:

R = 40 Ω/m, G = 400 μS/m, L = 0.2 μH/m, C = 0.5 nF/m

i) If the line operating at 10 MHz. Calculate the characteristic impedance Z_0 and velocity v_p.

ii) After how many meters will the voltage drop by 30 dB in the line?

Sol:

Given,

R = 40 Ω/m, G = 400 μS/m, L = 0.2 μH/m, C = 0.5 nF/m, f = 10 MHz

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= \sqrt{\frac{40 + j2\pi \times 10 \times 10^6 \times 0.2 \times 10^{-6}}{400 \times 10^{-6} + j2\pi \times 10 \times 10^6 \times 0.5 \times 10^{-9}}}$$

$$= \sqrt{\frac{41.92\angle 17.44°}{0.0314\angle 89.27°}}$$

$$= \sqrt{1335.03\angle -71.83°}$$

$$= 36.53\angle -35.92° = 29.5 - j21.4\ \Omega$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{(40 + j2\pi \times 10 \times 10^6 \times 0.2 \times 10^{-6})(400 \times 10^{-6} + j2\pi \times 10 \times 10^6 \times 0.5 \times 10^{-9})}$$

$$= \sqrt{(41.92\angle 17.44°)(0.0314\angle 89.27°)}$$

$$= 1.147\angle 53.35° = 0.68 + j0.92\ /m$$

β = 0.92 rad/m, α = 0.68 Np/m

v_p = Phase velocity $= \frac{\omega}{\beta} = \frac{2\pi \times 10 \times 10^6}{0.92} = 6.8 \times 10^7$ m/s

Attenuation constant (α) = 0.68 Np/m

We know that,

1 Np = 8.686 dB

$\therefore$ α = 0.68 × 8.686 = 5.9 dB/m

αl = 30 dB (Given)

$$l = \frac{30}{5.9} = 5.08 m$$

Rack Your Brain

What are the properties of infinite lines?

INPUT IMPEDANCE OF TRANSMISSION LINE

Consider a transmission line of length l, characterised by γ and Z_0 connected to a load Z_L as shown below.

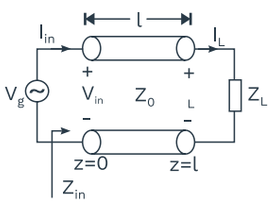


Fig. 6.4 Input Impedance Due to a Line Terminated by a Load

We Know that, transmission line equation:

$$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad ...(i)$$

$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad ...(ii)$$

$$Z_0 = \frac{V_0^+}{I_0^+} = \frac{-V_0^-}{I_0^-} \quad \text{...(iii)}$$

$$I_s(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} + \frac{-V_0^+}{Z_0} e^{\gamma z} \quad \text{...(iv)}$$

$$\text{At } z = 0,\ V_s(z=0) = V_0^+ + V_0^- = V_{in} \quad \text{...(v)}$$

$$\text{At } z = 0,\ V_s(z=0) = I_0^+ + I_0^- = I_{in}$$

and $V_0^+ - V_0^- = I_{in} Z_0$ ...(vi)

Adding equations v) and vi), we get:

$$2V_0^+ = V_{in} + I_{in} Z_0$$

$$V_0^+ = \frac{V_{in} + I_{in} Z_0}{2}$$

Subtracting equations v) and vi), we get:

$$2V_0^- = V_{in} - I_{in} Z_0$$

$$V_0^- = \frac{V_{in} - I_{in} Z_0}{2}$$

At $z = l$,

$$V_s(z = l) = V_0^+ e^{-\gamma l} + V_0^- e^{+\gamma l} = V_L \quad \text{...(vii)}$$

$$I_s(z = l) = \frac{V_0^+}{Z_0} e^{-\gamma l} - \frac{V_0^-}{Z_0} e^{+\gamma l} = I_L$$

Or $V_0^+ e^{-\gamma l} - V_0^- e^{+\gamma l} = I_L z_0$ ...(viii)

Adding equations vii) and viii), we get:

$$2V_0^+ e^{-\gamma l} = V_L + I_L Z_0$$

$$V_0^+ = \left(\frac{V_L + I_L Z_0}{2}\right) e^{\gamma l} = \left(\frac{I_L Z_L + I_L Z_0}{2}\right) e^{\gamma l}$$

Subtracting equations vii) & viii), we get:

$$2V_0^- e^{+\gamma l} = V_L - I_L Z_0$$

$$V_0^- = \left(\frac{I_L V_L - I_L Z_0}{2}\right) e^{-\gamma l}$$

Input impedance, $Z_{in} = \frac{V_{in}}{I_{in}}$

$$Z_{in} = \frac{V_0^+ + V_0^-}{\frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}}$$

Rack Your Brain

How to rectify the interference problem?

$$= Z_0\left[\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-}\right]$$

$$Z_{in} = Z_0\left[\frac{\left(\frac{I_L Z_L + I_L Z_0}{2}\right)e^{\gamma l} + \left(\frac{I_L Z_L - I_L Z_0}{2}\right)e^{-\gamma l}}{\left(\frac{I_L Z_L + I_L Z_0}{2}\right)e^{\gamma l} - \left(\frac{I_L Z_L - I_L Z_0}{2}\right)e^{-\gamma l}}\right]$$

$$= Z_0\left[\frac{Z_L\left(\frac{e^{\gamma l} + e^{-\gamma l}}{2}\right) + Z_0\left(\frac{e^{\gamma l} - e^{-\gamma l}}{2}\right)}{Z_0\left(\frac{e^{\gamma l} + e^{-\gamma l}}{2}\right) + Z_L\left(\frac{e^{\gamma l} - e^{-\gamma l}}{2}\right)}\right]$$

$$= Z_0\left[\frac{Z_L \cosh\gamma l + Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l + Z_L \sinh\gamma l}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l}\right]$$

This results in a lossy transmission line $(\gamma = \alpha + j\beta)$.

For lossless transmission line $\alpha = 0, \gamma = j\beta$.

$$\tanh(\gamma l) = \frac{\left(e^{\gamma l} - e^{-\gamma l}\right)}{\left(e^{\gamma l} + e^{-\gamma l}\right)}$$

$$= \frac{\left(e^{j\beta l} - e^{-j\beta l}\right)}{\left(e^{j\beta l} + e^{-j\beta l}\right)}$$

$$= \frac{j\left(\frac{e^{j\beta l} - e^{-j\beta l}}{2j}\right)}{\left(\frac{e^{j\beta l} + e^{-j\beta l}}{2}\right)}$$

$$= \frac{j\sin\beta l}{\cos\beta l} = j\tan\beta l$$

$$\therefore \quad Z_{in} = Z_0\left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}\right]$$

This results in a lossless transmission line.
Where, βl = electrical length (radian)

Rack Your Brain

What is a distortionless line?

Example 6: A lossless transmission line uses a dielectric insulating material with $\epsilon_r = 4$ if its line capacitance is C=10pF/m, Find

i) Phase velocity.
ii) Line inductance.
iii) Characteristics impedance

Sol:

Given, $\epsilon_r = 4, C = 10\,pF/m$

i) $v_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3\times10^8}{\sqrt{4}} = 1.5\times10^8 m/sec$

ii) $v_p = \frac{1}{\sqrt{LC}}$

$$L = \frac{1}{Cv_p^{\,2}} = \frac{1}{10\times10^{-12}\times\left(1.5\times10^8\right)^2}$$

$L = 4.4\ \mu H/m$

iii) $Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{4.44\times10^{-6}}{10\times10^{-12}}} = 666.33\ \Omega$

Grey Matter Alert!

i) Characteristic impedance of the co-axial cable is:

$$Z_0 = 60\sqrt{\frac{\mu_r}{\epsilon_r}}\ln\left(\frac{b}{a}\right)$$

ii) Characteristic impedance of the parallel line is:

$$Z_0 = 120\sqrt{\frac{\mu_r}{\epsilon_r}}\ln\left(\frac{D}{r}\right)$$

Input impedance for different lengths:

We know that, input impedance of transmission line is:

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_o\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

Case-1:

$$l = \frac{\lambda}{8},\quad \beta l = \frac{2\pi}{\lambda}\times\frac{\lambda}{8} = \frac{\pi}{4}$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_o\tan\frac{\pi}{4}}{Z_0 + jZ_L\tan\frac{\pi}{4}}\right]$$

$$= Z_0\left[\frac{Z_L + jZ_o}{Z_0 + jZ_L}\right]$$

$$|Z_{in}| = Z_0\frac{\sqrt{Z_L^2 + Z_0^2}}{\sqrt{Z_0^2 + Z_L^2}}$$

$$|Z_{in}| = Z_0$$

Case-2:

$$l = (2n-1)\frac{\lambda}{4}, n = 1,2,3,\ldots\ldots\left(\begin{matrix}\text{quarter - wave}\\ \text{transformer}\end{matrix}\right)$$

$$n = 1,\ l = \frac{\lambda}{4}, \beta l = \frac{2\pi}{\lambda}\times\frac{\lambda}{4} = \frac{\pi}{2}$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$= Z_0\left[\frac{\frac{Z_L}{\tan\frac{\pi}{2}} + jZ_o}{\frac{Z_0}{\tan\frac{\pi}{2}} + jZ_L}\right]$$

$$= \frac{Z_0 \times jZ_0}{jZ_L}$$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

$$|Z_{in}| = \frac{Z_0^2}{Z_L}$$

If Z_L = open circuit; Z_{in} = short circuit
If Z_L = short circuit; Z_{in} = open circuit
If Z_L = capacitive; Z_{in} = Inductive
If Z_L = Inductive; Z_{in} = capacitive

That is why $\frac{\lambda}{4}$ length transmission line is called an impedance inversion device.

Case-3:

$$l = \frac{n\lambda}{2} \qquad n = 1, 2, 3.....$$

$$n = 1,\ l = \frac{\lambda}{2}\ \left(\text{Half wavelength}\right)$$

$$\beta l = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{2} = \pi$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\pi}{Z_0 + jZ_L\tan\pi}\right]$$

$$Z_{in} = Z_L$$

If the length of the transmission line is integer multiple of $\frac{\lambda}{2}$, load impedance is appearing as input impedance without any change.

Previous Years' Question

A co-axial cable with an inner diameter of 1mm and an outer diameter of 2.4 mm is filled with a dielectric of relative permittivity of 10.89. Given,
$\mu_0 = 4\pi \times 10^{-7}$ H/m
$\varepsilon_0 = \frac{10^{-9}}{36\pi}$ F / m, the characteristic impedance of the cable is:

a) 330 Ω **b)** 100 Ω
c) 143.3 Ω **d)** 43.3 Ω

Input impedance for different load conditions:

Case-1:

$Z_L = Z_0$ (Matched Condition)

We know that,

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_0 + jZ_0\tan\beta l}{Z_0 + jZ_0\tan\beta l}\right]$$

$$Z_{in} = Z_0$$

Case-2:

$Z_L = 0$ (short circuited)

We know that,

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$= Z_0\left[\frac{0 + jZ_0\tan\beta l}{Z_0 + 0}\right]$$

$$Z_{in} = jZ_0\tan\beta l = Z_{SC}$$

When the load is short-circuited, the input impedance of the transmission line is denoted by Z_{SC}.

1) $0 < l < \frac{\lambda}{4}$ then $0 < \beta l < \frac{\pi}{2}$ $\left(1^{st}\text{ quadrant}\right)$

$$Z_{SC} = jZ_0\tan\beta l = \underset{\text{(Purely inductive)}}{jZ_0K} \quad \{\text{when } K=+^{Ve} \text{ Value}\}$$

2) $\frac{\lambda}{4} < l < \frac{\lambda}{2}$ then $\frac{\pi}{2} < \beta l < \pi$ $\left(2^{nd}\text{ quadrant}\right)$

$$Z_{SC} = jZ_0\tan\beta l = \underset{\text{(Purely capacitive)}}{jZ_0(-K)} \quad \{\text{when } K=+^{Ve} \text{ Value}\}$$

3) $\frac{\lambda}{2} < l < \frac{3\lambda}{4}$, $\pi < \beta l < \frac{3\pi}{2}$ $\left(3^{rd}\text{ quadrant}\right)$

$$Z_{SC} = jZ_0\tan\beta l = \underset{\text{(Purely inductive)}}{jZ_0(K)} \quad \{\text{when } K=+^{Ve} \text{ Value}\}$$

4) $\frac{3\lambda}{4} < l < \lambda$, $\frac{3\pi}{2} < \beta l < 2\pi$ $\left(4^{th}\text{ quadrant}\right)$

$$Z_{SC} = jZ_0\tan\beta l = \underset{\text{(Purely capacitive)}}{jZ_0(-K)} \quad \{\text{when } K=+^{Ve} \text{ Value}\}$$

Grey Matter Alert!

The lossless line is the distortion less but vice versa is not true.

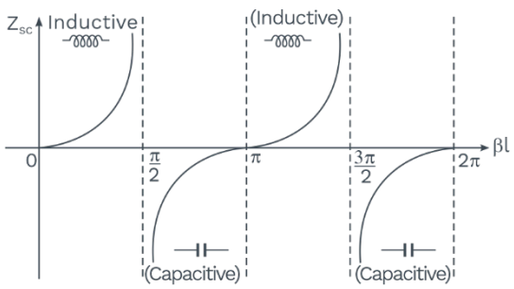


Fig. 6.5 Input Impedance of a Lossless Line when Load is Shorted

Case-3:

$Z_L = \infty$ (open circuited)

We know that,

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$$

$$= Z_0\left[\frac{1+\frac{jZ_0\tan\beta l}{Z_L}}{\frac{Z_0}{Z_L}+j\tan\beta l}\right]$$

$$= Z_0\left[\frac{1+0}{0+j\tan\beta l}\right]$$

$$= -jZ_0\cot\beta l$$

$$Z_{in} = jZ_0\cot\beta l = Z_{OC}$$

This results in the input impedance of the transmission line when the load is open-circuited.

1) $0 < l < \frac{\lambda}{4}$, $0 < \beta l < \frac{\pi}{2}$

$Z_{OC} = -jZ_0\cot\beta l = -jZ_0K$ (Capacitive)

2) $\frac{\lambda}{4} < l < \frac{\lambda}{2}$ then $\frac{\pi}{2} < \beta l < \pi$

$Z_{OC} = -jZ_0\cot\beta l = jZ_0K$ (Inducitive)

3) $\frac{\lambda}{2} < l < \frac{3\lambda}{4}$, $\pi < \beta l < \frac{3\pi}{2}$

$Z_{OC} = -jZ_0\cot\beta l = -jZ_0K$ (Capacitive)

4) $\frac{3\lambda}{4} < l < \lambda$, $\frac{3\pi}{2} < \beta l < 2\pi$

$Z_{OC} = -jZ_0\cot\beta l = jZ_0K$ (Inducitive)

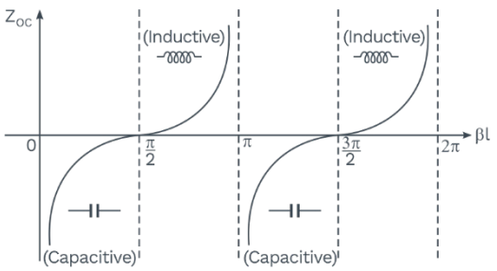


Fig. 6.6 Input Impedance of a Lossless Line when Open

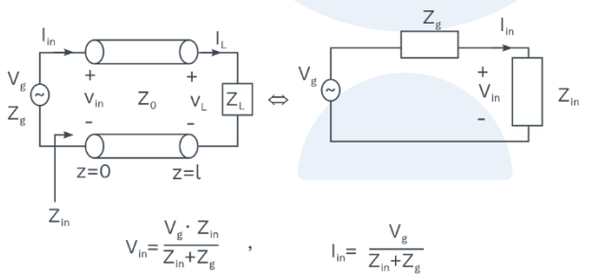
$$Z_{SC} = jZ_0 \tan \beta l$$

$$Z_{OC} = -jZ_0 \cot \beta l$$

$$Z_0 = \sqrt{Z_{OC} Z_{SC}}$$

Characteristics impedance is the geometric mean of open-circuit impedance Z_{OC} and short-circuit impedance Z_{SC}

Note:



REFLECTION COEFFICIENT AT LOAD

The reflection coefficient is the ratio of the reflected wave to the incident wave.

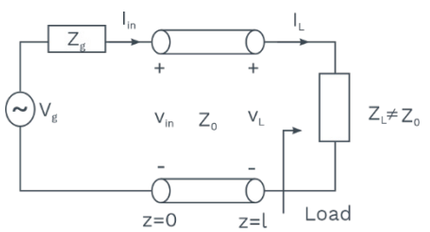


Fig. 6.7 Reflection Coefficients Due to Transmission Line Terminated by a Load

$$\text{Reflection coefficent } (\Gamma_V) = \frac{\text{Reflected voltage wave}}{\text{Incident voltage wave}}$$

We know that,

$$V_S(z) = \underset{\text{Incident}}{V_0^+} e^{-\gamma z} + \underset{\text{Reflected}}{V_0^-} e^{\gamma z}$$

$$\therefore \Gamma_V = \Gamma_{load} = \frac{V_0^- e^{\gamma l}}{V_0^+ e^{-\gamma l}}$$

$$\Gamma_{load} = \frac{V_0^-}{V_0^+} e^{2\gamma l} \quad \ldots(i)$$

$$V_S(l) = V_0^+ e^{-\gamma l} + V_0^- e^{\gamma l} = V_L \quad \ldots(ii)$$

$$I_S(l) = \frac{V_0^+}{Z_0} e^{-\gamma l} - \frac{V_0^-}{Z_0} e^{\gamma l} = I_L$$

$$V_0^+ e^{-\gamma l} - V_0^- e^{\gamma l} = I_L Z_0 \quad \ldots(iii)$$

Adding equations ii) and iii), we get:

$$2V_0^+ e^{-\gamma l} = V_L + I_L Z_0$$

$$V_0^+ = \frac{V_L + I_L Z_0}{2e^{-\gamma l}} = \frac{I_L Z_L + I_L Z_0}{2e^{-\gamma l}}$$

$$V_0^+ = \frac{I_L (Z_L + Z_0) e^{\gamma l}}{2}$$

Subtracting equations ii) and iii), we get:

$$V_0^- = \frac{I_L (Z_L - Z_0) e^{-\gamma l}}{2}$$

From equation i), we get:

$$\Gamma_{load} = \frac{\frac{I_L}{2}(Z_L - Z_0) e^{+\gamma l}}{\frac{I_L}{2}(Z_L + Z_0) e^{+\gamma l}}$$

$$\Gamma_{load} = \frac{(Z_L - Z_0)}{(Z_L + Z_0)}$$

Definitions

Voltage reflection coefficient:
The voltage reflection coefficient at any point on the line is the ratio of the reflected voltage wave to the incident wave.

Rack Your Brain

What is the reflection coefficient?

Reflection coefficient at any distance l′ from load:

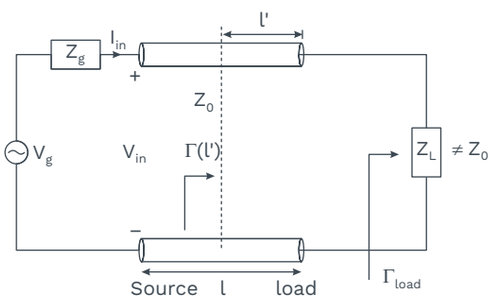


Fig. 6.8 Reflection Coefficient at Any Distance l′ from Load Due to Line Terminated by a Load

$$V_S(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$V_S(z = l - l') = V_0^+ e^{-\gamma(l-l')} + V_0^- e^{\gamma(l-l')}$$

$$= \underbrace{V_0^+ e^{-\gamma l} e^{\gamma l'}}_{\text{forward}} + \underbrace{V_0^- e^{\gamma l} e^{-\gamma l'}}_{\text{Backward}}$$

Reflection coefficient,

$$\Gamma(l') = \frac{V_0^- e^{\gamma l} e^{-\gamma l'}}{V_0^+ e^{-\gamma l} e^{\gamma l'}}$$

$$\Gamma(l') = \left[\frac{V_0^-}{V_0^+} e^{+2\gamma l} . e^{-2\gamma l'}\right]$$

$$\Gamma(l') = \frac{(Z_L - Z_0)}{(Z_L + Z_0)} e^{-2\gamma l'}$$

Where, $\Gamma_{load} = \dfrac{Z_L - Z_0}{Z_L + Z_0}$

Current reflection coefficient,

$$\Gamma_I = \frac{\text{Reflected current wave}}{\text{Incident current wave}}$$

$$= \frac{\frac{-V_0^-}{Z_0} e^{\gamma l}}{\frac{V_0^+}{Z_0} e^{-\gamma l}}$$

$$= -\left(\frac{V_0^-}{V_0^+}\right) e^{2\gamma l}$$

$$\Gamma_I = -\left(\frac{Z_L - Z_0}{Z_L + Z_0}\right)$$

or $\Gamma_I = -\Gamma_V$

For matched condition, $Z_L = Z_0$

$$\Gamma_V = \Gamma_I = 0$$

$$\Gamma(l') = \Gamma_{load}\ e^{-2\gamma l'}$$

where, $\Gamma_{(load)} = \dfrac{Z_L - Z_0}{Z_L + Z_0}$

$$\Gamma(l') = \Gamma_{load}\ e^{-2(\alpha + j\beta)l'}$$

$$\Gamma(l') = \underbrace{\Gamma_{load}\ e^{-2\alpha l'}}_{\text{Magnitude}} \underbrace{e^{-j2\beta l'}}_{\text{Phase}}$$

Definitions

Current reflection coefficient: The current reflection coefficient at any point on the line is the negative of the voltage reflection coefficient at that point.

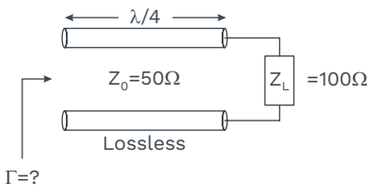
$$\Gamma_I = -\Gamma_L$$

For lossless transmission line, $\alpha = 0$

$$\Gamma(l') = \Gamma_{load}\, e^{-j2\beta l'}$$

Example 7:

For the lossless transmission line shown below. Determine reflection coefficient Γ.



Sol:

Given, $Z_L = 100\Omega$, $Z_0 = 50\Omega$

$l' = \lambda/4$

$$\beta l' = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$$

$$\Gamma_{load} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 50}{100 + 50} = \frac{1}{3}$$

$$\Gamma(l') = \Gamma_{load}\, e^{-j2\beta l'}$$

$\frac{\lambda}{4}$ distance from load,

$$\Gamma\left(l' = \frac{\lambda}{4}\right) = \Gamma_{load} e^{-j2\frac{\pi}{2}}$$

$$= \frac{1}{3} e^{-j\pi}$$

$$= \frac{1}{3}[\cos\pi - j\sin\pi]$$

$$\Gamma\left(l' = \frac{\lambda}{4}\right) = -\frac{1}{3}$$

Rack Your Brain

What is the disadvantage of reflection?

Reflection coefficient for complex load ($Z_L = R + jX$) at any distance l' from load:

$$\Gamma_{load} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(R + jX) - Z_0}{(R + jX) + Z_0}$$

$$\Gamma_{load} = \sqrt{\frac{(R - Z_0)^2 + X^2}{(R + Z_0)^2 + X^2}} \angle \left[\tan^{-1}\left(\frac{X}{R - Z_0}\right) - \tan^{-1}\left(\frac{X}{R + Z_0}\right)\right]$$

$\Gamma_{load} = |\Gamma_{load}|e^{j\phi}$

At any distance l′ from load,

$\Gamma(l') = |\Gamma_{load}|e^{j\phi}e^{-2\gamma l'}$

$\Gamma(l') = |\Gamma_{load}|e^{j\phi}e^{-2\alpha l'}e^{-j2\beta l'}$

$\Gamma(l') = |\Gamma_{load}|e^{-2\alpha l'}e^{j(\phi-2\beta l')}$

For lossless transmission line, $\alpha = 0$

$\Gamma(l') = |\Gamma_{load}|e^{j(\phi-2\beta l')}$

where, ϕ = phase of reflection coefficient at load (complex)

STANDING WAVE RATIO

The ratio of maximum voltage to minimum voltage and its magnitude is called voltage standing wave ratio (VSWR).

i.e. $VSWR = \left|\frac{V_{max}}{V_{min}}\right| = \frac{1+|\Gamma_V|}{1-|\Gamma_V|}$

VSWR at load,

$VSWR = \frac{1+|\Gamma_{load}|}{1-|\Gamma_{load}|}$

VSWR at any distance l' from the load,

$VSWR = \frac{1+|\Gamma(l')|}{1-|\Gamma(l')|}$

We know that, $\Gamma(l') = \Gamma_{load}\, e^{-2\gamma l'}$

$= \Gamma_{load}\, e^{-2\alpha l'}\, e^{-j2\beta l'}$

$|\Gamma(l')| = |\Gamma_{load}\, e^{-2\alpha l'}|$

$\therefore\ VSWR = \frac{1+|\Gamma_{load}e^{-2\alpha l'}|}{1-|\Gamma_{load}e^{-2\alpha l'}|}$

This results in a lossy transmission line.

For lossless transmission line, $\alpha = 0$

$VSWR = \frac{1+|\Gamma_{load}|}{1-|\Gamma_{load}|}$

If the transmission line is lossless, then VSWR remains constant.

The ratio of maximum current to minimum current and its magnitude is called as current standing wave ratio (CSWR).

i.e. $CSWR = \left|\frac{I_{max}}{I_{min}}\right| = \frac{1+|\Gamma_I|}{1-|\Gamma_I|}$

Definitions

It is the ratio of the maximum voltage of the standing wave to the minimum voltage of the standing wave.

We know that, $\Gamma_I = -\Gamma_V$

$\therefore$ CSWR $= \dfrac{1+|\Gamma_V|}{1-|\Gamma_V|}$

$\therefore$ VSWR = CSWR

Standing wave pattern:

Case-1: $Z_L > Z_0$ (Z_L = resistive)

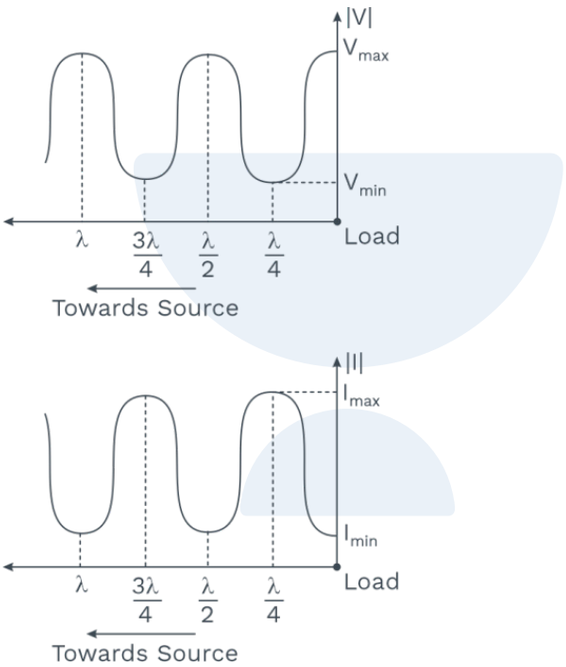


Fig. 6.9 Voltage and Current Wave Patterns on a Lossless Line Terminated by a Resistive Load

1) Distance between two successive maxima or minima is $\lambda/2$.

2) If $Z_L > Z_0$, VSWR $= \dfrac{Z_L}{Z_0}$

Rack Your Brain

What are the standing waves?

Case-2: $Z_0 > Z_L$ (Z_L = resistive)

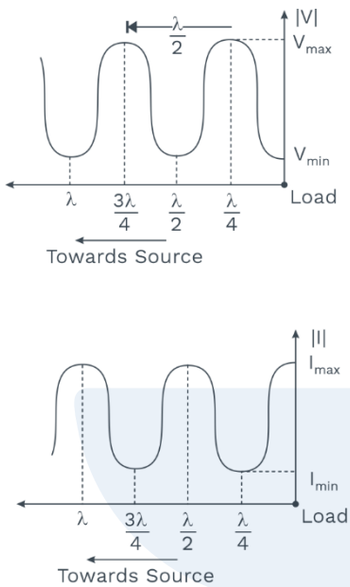


Fig. 6.10 Voltage and Current Wave Patterns on a Lossless Line Terminated by a Resistive Load

Standing wave ratio, $Z_0 > Z_L$; $VSWR = \frac{Z_0}{Z_L}$

Reflection coefficient and vswr for different load:

1) Load is short-circuited, Z_L=0

$$\Gamma_{load} = \frac{Z_L - Z_O}{Z_L + Z_O} = -1$$

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \infty$$

2) Load is open-circuited, Z_L=∞

$$\Gamma_{load} = \frac{1 - \frac{Z_0}{Z_L}}{1 + \frac{Z_0}{Z_L}} = 1, \quad SWR=\infty$$

3) Load is equal to characteristics impedance

$$Z_L = Z_0$$

$$\Gamma_{load} = 0, \quad SWR=1$$

Range of reflection coefficient Γ is:

Previous Years' Question

A 50 Ohm lossless transmission line has a pure reactance of j100 Ohms as its load. The VSWR in the line is:

a) $\frac{1}{2}$ (Half)
b) 2 (Two)
c) 4 (Four)
d) ∞ (Infinity)

$-1 \leq G \leq 1$

Range of SWR is:

$1 \leq S \leq \infty$

Location of maxima and minima:

If, $Z_L > Z_O$

$$l'_{max} = \frac{\phi + 2n\pi}{2\beta} \quad n=0, 1, 2, \ldots \text{ if } \phi \geq 0$$

$$= 1, 2, 3 \ldots \text{ if } \phi < 0$$

ϕ = Phase of reflection coefficient (Z_L is complex)

$$l'_{min} = l'_{max} - \frac{\lambda}{4}; \quad \text{if } l'_{max} \geq \frac{\lambda}{4}$$

$$l'_{min} = l'_{max} + \frac{\lambda}{4}; \quad \text{if } l'_{max} < \frac{\lambda}{4}$$

Rack Your Brain

What is the minimum value of VSWR?

IMPEDANCE MATCHING DEVICES

There are different impedance matching devices used in the transmission lines.

Impedance matching using a quarter-wave transformer ($\lambda/4$ transformer):

When the impedance of a load is the same as the characteristic impedance of the transmission line, there is no reflected wave, all the forward going power is dissipated in the load. There are various ways to achieve this impedance matching like a quarter-wave transformer, single-stub matching, double-stub matching, etc.

Quarter-wave transformer, like low-frequency transformers, changes the impedance of the load to another value so that matching is possible.

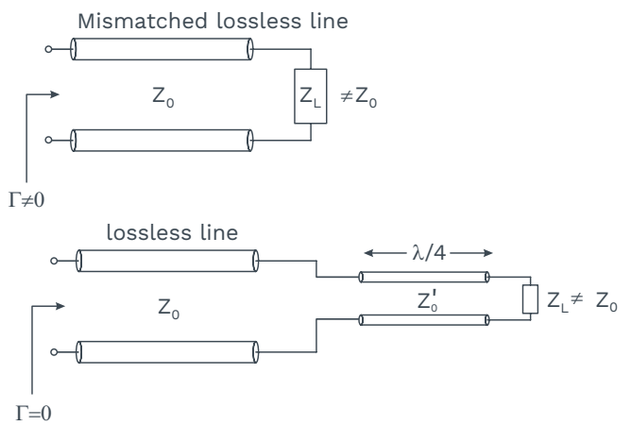


Fig. 6.11 Load Matching Using a $\lambda/4$ Transformer

We know that input impedance of transmission line is:

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}\right]$$

Load impedance of lossless line will be input impedance of the $\frac{\lambda}{4}$ line.

$$\beta l = \frac{\pi}{2}$$

$$Z_{in} = \frac{Z_0'^2}{Z_L}$$

For $\Gamma = 0, Z_{in} = Z_0$

$$Z_0 = \frac{Z_0'^2}{Z_L}$$

$$Z_0'^2 = Z_0 Z_L$$

$$Z_0' = \sqrt{Z_0 Z_L}$$

Single stub impedance matching:

Single-stub method for impedance matching, an arbitrary load impedance can be matched to a transmission line by placing a single short-circuited stub in parallel with the line at a suitable location, the input admittance at AA' is looking towards the load without the stub.

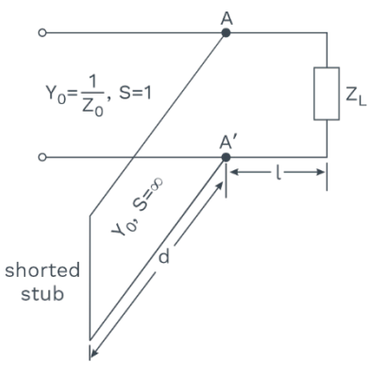


Fig. 6.12 Load Matching with a Single-Stub Matching

As we intend that $Z_{in} = Z_0$, that is, $Z_{in} = 1$ or $Y_{in} = 1$ at point A on the line.

If $Y = 1 + jb$ and $Y_s = -jb$

Therefore, $Y_{in} = Y + Y_s$

$= 1 + jb - jb$

$= 1 + j0$

Rack Your Brain

What are the applications of quarter-wave lines?

Advantages of single-stub matching:

- The single-stub matching technique is superior to the quarter wavelength transformer as it makes use of only one type of transmission line for the mainline as well as the stub.
- This technique also, in principle, is capable of matching any complex load to the characteristic impedance/admittance.
- The single stub matching technique is quite popular in matching fixed impedance at microwave frequencies.

Disadvantages of single-stub matching:

The single-stub matching technique has overcome the drawback of the quarter wavelength transformer technique, it is still not suitable for matching variable impedances. A change in load impedance results in a change in the length as well as the location of the stub. Even if changing the length of a stub is a simpler task, changing the location of a stub may not be easy in certain transmission line configurations. For example, if the transmission line is a co-axial cable, the connection of a stub would need drilling of a hole in the outer conductor.

Double stub impedance matching:

Another possible method of impedance matching is to use two stubs in which the locations of the stub are arbitrary, there are two stub lengths required to furnish the adjustments. The spacing is frequently made 1/4. This is called double stub matching.

Double stub matching is preferred over single stubs due to the following disadvantages of the single stub.

1) The single-stub matching system is based on the measurement of voltage minimum. Hence for the co-axial line, it is very difficult to get such voltage minimum without using a slotted line section.

Rack Your Brain

What is a single stub matching line?

Grey Matter Alert!

Shunt stub matching:
A short-circuited stub is placed at a distance $l_S = \frac{\lambda}{2\pi}\tan^{-1}\sqrt{\frac{Z_L}{Z_0}}$ from the load.

2) Single-stub matching is useful for a fixed frequency. So as the frequency changes, the location of a single stub has to be changed.

Example 8:

Find the input impedance of a short-circuited co-axial transmission line if $Z_0 = 65 + j38\ \Omega$, γ=0.7+j2.5 /m , l = 0.8m.

Sol:

Given,

$Z_0 = 65 + j38\ \Omega$,

$\gamma = 0.7+j2.5/m$

$l = 0.8m$

$$Z_{in} = Z_0\left[\frac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l}\right]$$

For short circuit, $Z_L = 0$

$$Z_{in} = Z_0 \tanh(\gamma l)$$

$$\gamma l = (0.7 + j2.5) \times 0.8 = 0.56 + j2 = x + jy$$

$\sinh(\gamma l)=\sinh(x+jy)=\sinh(x)\cos y+j\cosh(x)\sin y$

$$= -0.245 + j1.056$$

$$\cosh\gamma l = \cosh(x + jy) = \cosh(x)\cos y + j\sinh(x)\sin(y)$$

$$= -0.483 + j0.536$$

$$Z_{in} = \frac{(65 + j38)(-0.245 + j1.056)}{(-0.483 + j0.536)}$$

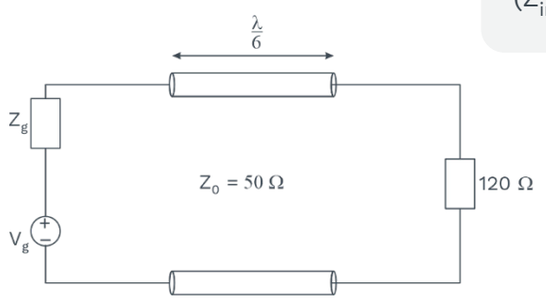
$$Z_{in} = (113.1 + j2.67)\ \Omega$$

Example 9:

Refer to the lossless line shown in the figure.

i) Find Γ and standing wave ratio (S).

ii) Determine Z_{in} at the generator.



Grey Matter Alert!

The magnitude of Z_{in} for $\frac{\lambda}{8}$ line and l = ∞ length line are equal but the phase is different. So, l = ∞ and $l = \frac{\lambda}{8}$ input impedance (Z_{in}) is not equal.

Sol:

i) Given $Z_0 = 50\ \Omega,\ Z_L = 120\ \Omega$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{120 - 50}{120 + 50} = 0.411$$

$$Z_L > Z_0$$

$$S = \frac{Z_L}{Z_0} = \frac{120}{50} = 2.4$$

ii) $$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan(\beta l)}{Z_0 + jZ_L\tan(\beta l)}\right]$$

$$Z_{in} = 50\left[\frac{120 + j50\tan\left(\frac{\pi}{3}\right)}{50 + j120\tan\left(\frac{\pi}{3}\right)}\right] \quad \left\{\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}\right\}$$

$$Z_{in} = 50\left[\frac{120 + j86.6}{50 + j207.84}\right]$$

$$= 26.26 - j22.55$$

$$= 34.61\angle 40.65^\circ\ \Omega$$

Example 10:

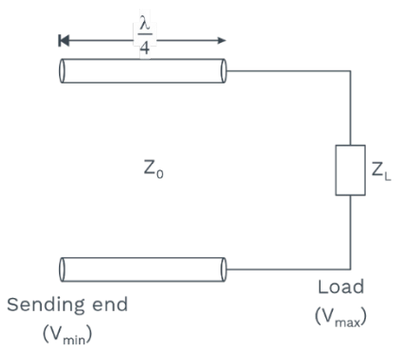
A quarter-wave lossless 100 Ω line is terminated by a load Z_L = 210 Ω. If the voltage at the receiving end is 80 V, what is the voltage at sending end?

Sol:

Given, $Z_0 = 100\ \Omega, Z_L = 210\ \Omega$

V_{max} = 80 V

If the load is resistive ($Z_L > Z_0$), V_{max} occurs at load.



$$VSWR = \frac{V_{max}}{V_{min}}$$

$$V_{min} = \frac{V_{max}}{VSWR}$$

$$= \frac{V_{max} Z_0}{Z_L} = \frac{80 \times 100}{210} = 38.1 \text{ V}$$

Grey Matter Alert!

Characteristics impedance is the geometric mean of short-circuited and open-circuited line input impedance.

$$Z_0 = \sqrt{Z_{SC} Z_{OC}}$$

Example 11:

A 500-Ω lossless line has $V_L = 10e^{j25°}$ V; $Z_L = 50e^{j30°}$.

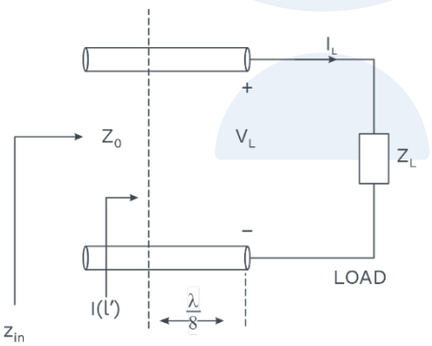
Find the current at $\frac{\lambda}{8}$ from the load.

Sol:

Given,

$$V_L = 10e^{j25°}, \quad Z_L = 50e^{j30°}$$

$$I_L = \frac{V_L}{Z_L} = \frac{10e^{j25°}}{50e^{j30°}} = 0.2e^{-j5°}$$



$$I(l') = I_L e^{\gamma l'} = I_L e^{(\alpha + j\beta) l'}$$

$$= I_L e^{j\beta l'} \quad [\text{for lossless line } \alpha = 0]$$

$$= I_L e^{j\frac{2\pi}{\lambda} \times \frac{\lambda}{8}}$$

$$= I_L e^{j\frac{\pi}{4}}$$

$$= I_L e^{j45°}$$

$$= 0.2e^{-j5°} \; e^{j45°}$$

$$= 0.2e^{j40°} \text{ A}$$

Example 12:
A co-axial line 5.6 m long has distributed parameters R = 6.5 Ω/m, L = 3.4 μH/m, G = 8.4 mS/m and C = 21.5 pF/m.
if the line operates at 2 MHz, calculate the characteristics of impedance and the end-to-end propagation time delay.

Sol:
Given,

$R = 6.5\ \Omega/m,\ L = 3.4\ \mu H/m$

$G = 8.4\ mS/m,\ C = 21.5\ pF/m$

$l = 5.6\ m,\ f = 2\ MHz$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$= \sqrt{\frac{6.5 + j2\pi \times 2 \times 10^6 \times 3.4 \times 10^{-6}}{8.4 \times 10^{-3} + j2\pi \times 2 \times 10^6 \times 21.5 \times 10^{-12}}}$$

$$= \sqrt{\frac{6.5 + j42.704}{0.0084 + j2.7 \times 10^{-4}}}$$

$$= \sqrt{\frac{43.2\ \angle 81.345^\circ}{8.404 \times 10^{-3}\ \angle 1.84^\circ}} = \sqrt{5140.4\ \angle 79.505^\circ}$$

$$= 71.69\ \angle 39.75^\circ$$

$$Z_0 = (51.2 + j45.84)\ \Omega$$

Propagation time delay:

$$t = \frac{l}{v_p} = \frac{l}{\frac{\omega}{\beta}} = \frac{\beta l}{\omega}$$

$$= \sqrt{\left(6.5 + j2\pi \times 2 \times 10^6 \times 3.4 \times 10^{-6}\right)\left(8.4 \times 10^{-3} + j2\pi \times 2 \times 10^6 \times 21.5 \times 10^{-12}\right)}$$

$$= \sqrt{0.363\ \angle 83.18^\circ}$$

$$= 0.602\ \angle 41.59^\circ$$

$$= 0.45 + j0.39\ m^{-1}$$

$$\therefore \beta = 0.39$$

$$\therefore t = \frac{0.39 \times 5.6}{2\pi \times 2 \times 10^6} = 1.738 \times 10^{-7}\ sec$$

$$\therefore t = 0.1738\ \mu\ sec$$

Previous Years' Question

A load impedance, (200 + j0)Ω is to be matched to a 50Ω lossless transmission line by using a quarter-wave line transformer (QWT). The characteristic impedance of QWT required (in ohms) is _____.

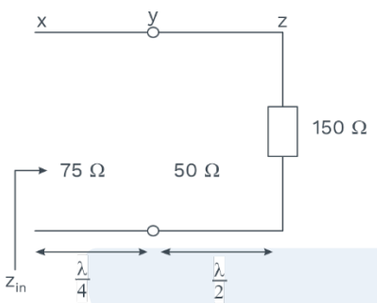
Example 13:

Two lines are cascaded, as shown in the figure. Determine:

i) The input impedance.

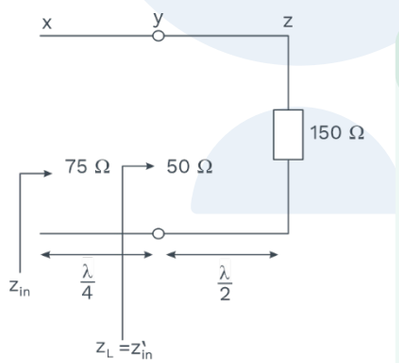
ii) The standing wave ratio for sections xy and yz.

iii) The reflection coefficient at z.



Sol:

i) The given figure is:



For $\frac{\lambda}{2}$ length transmission line:

$$Z'_{in} = Z_L = 150\,\Omega$$

For $\frac{\lambda}{4}$ length transmission line:

$$Z'_{in} = \frac{Z_0^2}{Z_L} = \frac{75^2}{150} = 37.5\,\Omega$$

ii) Standing waves ratio for yz section:

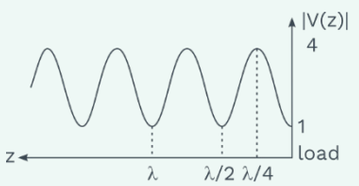
$$S_{yz} = \frac{Z_L}{Z_0} = \frac{150}{50} = 3$$

Standing wave ratio for xy section:

$$S_{xy} = \frac{Z_L}{Z_0} = \frac{150}{75} = 2$$

Previous Years' Question

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50Ω and resistive load is shown in the figure.



1) The value of the load resistance is:

a) 50Ω **b)** 200Ω

c) 12.5Ω **d)** 0

2. The reflection coefficient is given by:

a) −0.6 **b)** −1

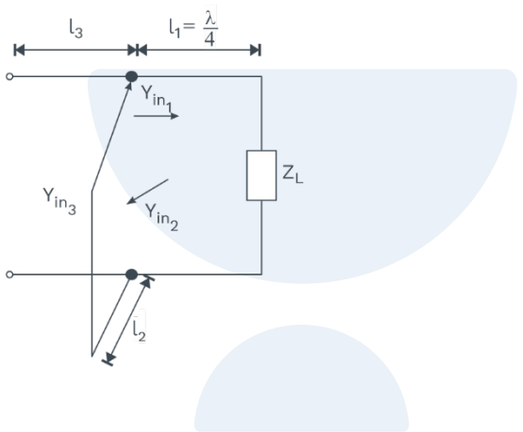
c) 0.6 **d)** 0.2

iii) Reflection coefficient at z,

$$\Gamma_{load} = \frac{Z_L - Z_O}{Z_L + Z_O} = \frac{150 - 50}{150 + 50} = \frac{1}{2} = 0.5$$

Example 14:

A section of the lossless transmission line is shunted across the mainline as in the figure. If $l_1 = \frac{\lambda}{4}$, $l_2 = \frac{\lambda}{8}$ and $l_3 = \frac{7\lambda}{8}$, find Y_{in_1}, Y_{in_2} and Y_{in_3}. Given that $Z_0 = 100\ \Omega, Z_L = 200 + j150\ \Omega$.



Sol:

Given,

$$l_1 = \frac{\lambda}{4}, l_2 = \frac{\lambda}{8}, l_3 = \frac{7\lambda}{8}$$

$$Z_0 = 100\ ,\ Z_L = (200 + j150)\ \Omega$$

$$Z_{in_1} = \frac{Z_0^2}{Z_L} = \frac{(100)^2}{200 + j150}$$

$$Y_{in_1} = \frac{1}{Z_{in_1}} = \frac{200 + j150}{(100)^2} = (20 + j15)\,mS$$

$$Z_{in_2} = Z_0\left[\frac{Z_L + jZ_0}{Z_0 + jZ_L}\right]$$

$$= jZ_0 \qquad \text{[Load short circuited]}$$

$$= j100\ \Omega$$

$$Y_{in_2} = \frac{1}{j100} = -j10\ mS$$

$$Y_L = Y_{in_1} + Y_{in_2}$$

Definitions

Return loss:

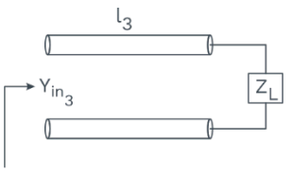
It is the ratio of the reflected power to the incident power in dB.

$$R_L = -20\log\Gamma$$

$= 20+j15-j10$

$= 20+j5 \quad mS$

$$Z_L = \frac{1}{Y_L} = \frac{1}{(20+j5)\times 10^{-3}} = (47.06 - j11.76)\ \Omega$$



$$\beta l_3 = \frac{2\pi}{\lambda}\times\frac{7\lambda}{8} = \frac{7\pi}{4} \Rightarrow \tan\left(\frac{7\pi}{4}\right) = -1$$

$$Z_{in_3} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l_3}{Z_0 + jZ_L\tan\beta l_3}\right]$$

$$= 100\left[\frac{47.06 - j11.76 + j100(-1)}{100 + j(47.06 - j11.76)\times(-1)}\right]$$

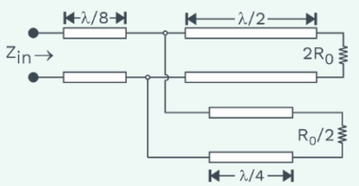
$$Y_{in_3} = \frac{100 - j(47.06 + j11.76)}{100(47.06 - 11.76 - j100)}$$

$$= \frac{88.24 - j47.06}{3530 - j10000}$$

$$Y_{in_3} = (6.408 + j6.36)\ mS$$

Previous Years' Question

All transmission line sections in the figure have a characteristic impedance R_0 + j0. The input impedance Z_{in} equals:



a) $\frac{2}{3}R_0$ **b)** R_0

c) $\frac{3}{2}R_0$ **d)** $2R_0$

SMITH CHART

1) It is the representation of the reflection coefficient in terms of normalised impedance.

2) It is used to determine input impedance, reflection coefficient, VSWR, location of maxima and minima, and admittance.

3) It is polar between the real part and the imaginary part of the reflection coefficient.

4) There are two families of circles:
- **i)** Constant resistance circle
- **ii)** Constant reactance circle.

5) Constant resistance and constant reactance circle are orthogonal to each other.

6) There are three scales on the outer of the smith chart:
- **i)** Angle or phase (in degree)

Definitions

Smith chart is a graphical technique used for calculating reflection coefficient and standing wave ratio for normalised load impedance.

ii) Distance in terms of wavelength towards the generator.
iii) Distance in terms of wavelength towards the load.

Assumption:
1) Lossless transmission line
2) Load that can be realised using transmission line is passive load.
Passive load→Positive resistive part.

Note: Normalised impedance, $Z_n = \frac{Z_L}{Z_0}$

If the load is A + jB.

$$\therefore Z_n = \frac{A + jB}{Z_0} = \frac{A}{Z_0} + j\frac{B}{Z_0}$$

$$Z_n = r + jx$$

Reflection coefficient,

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$= \frac{\frac{Z_L}{Z_0} - 1}{\frac{Z_L}{Z_0} + 1} = \frac{Z_n - 1}{Z_n + 1}$$

$$\Gamma = \frac{r + jx - 1}{r + jx + 1} = \frac{(r-1) + jx}{(r+1) + jx}$$

$$\Gamma = |\Gamma| e^{j\phi}$$

$$\Gamma = \frac{Z_n - 1}{Z_n + 1}$$

$$Z_n = \frac{1 + \Gamma}{1 - \Gamma} = \frac{1 + \Gamma_{real} + j\Gamma_{img}}{1 - \Gamma_{real} - j\Gamma_{img}}$$

$$r + jx = \frac{1 + \Gamma_{real} + j\Gamma_{img}}{1 - \Gamma_{real} - j\Gamma_{img}} \times \frac{(1 - \Gamma_{real}) + j\Gamma_{img}}{(1 - \Gamma_{real}) + j\Gamma_{img}}$$

$$= \frac{1 + \Gamma_{real} + j\Gamma_{img} - \Gamma_{real} - \Gamma_{real}^2 - j\Gamma_{img}\Gamma_{real} + j\Gamma_{img} - j\Gamma_{real}\Gamma_{img} - \Gamma_{img}^2}{(1 - \Gamma_{real})^2 + \Gamma_{img}^2}$$

$$r + jx = \frac{1 - \Gamma_{real}^2 - \Gamma_{img}^2 + j2\Gamma_{img}}{(1 - \Gamma_{real})^2 + \Gamma_{img}^2}$$

$$r = \frac{1 - \Gamma_{real}^2 - \Gamma_{img}^2}{(1 - \Gamma_{real})^2 + \Gamma_{img}^2}, \quad x = \frac{2\Gamma_{img}}{(1 - \Gamma_{real})^2 + \Gamma_{img}^2}$$

Rack Your Brain

What are the properties of the Smith chart?

After rearranging:

$$\left(\Gamma_{real} - \frac{r}{r+1}\right)^2 + \Gamma_{img}^2 = \left(\frac{1}{r+1}\right)^2$$

Constant resistance circle centre = $\left(\frac{r}{r+1}, 0\right)$, radius = $\frac{1}{r+1}$

Similarly, $(\Gamma_{real} - 1)^2 + \left(\Gamma_{img} - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$

Constant reactance circle centre $\left(1, \frac{1}{x}\right)$, radius= $\frac{1}{x}$.

Rack Your Brain

What is the significance of a circular diagram?

Assume:

i) If $Z_L = (50 + j100)\Omega$, $Z_0 = 25\Omega$, then $Z_n = (2 + j4)$

ii) If $Z_L = (75 - j100)\Omega$, $Z_0 = 25\Omega$, then $Z_n = (3 - j4)$

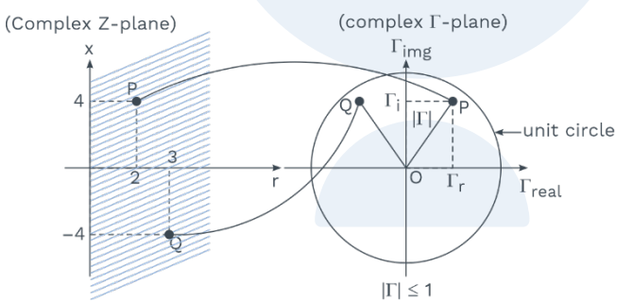


Fig. 6.13 Smith Chart in the Complex Γ Plane

Constant resistance circle:

$$\left(\Gamma_{real} - \frac{r}{r+1}\right)^2 + \Gamma_{img}^2 = \left(\frac{1}{r+1}\right)^2$$

centre $\left(\frac{r}{r+1}, 0\right)$, radius $\left(\frac{1}{r+1}\right)$

r	Centre	Radius
0	(0,0)	1
1	(0.5,0)	0.5
∞	(1,0)	0
0.5	(0.33, 0)	0.66

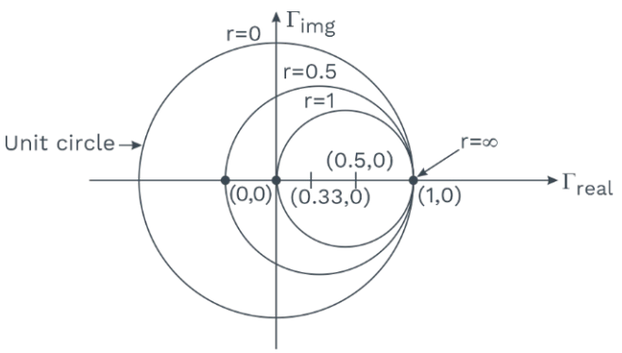


Fig. 6.14 Typical r-Circles for r = 0, 0.5, 1, ∞.

1) For $1 \le r \le \infty$, all the circles lie in the 1st and 4th quadrant.
2) All circles pass through the point (1, 0).
3) As the value of r increases centre shifts towards the right.
4) $\Gamma_{real} = 1$ is tangent to all the circles.

Constant reactance circle:

$$\left(\Gamma_{real} - 1\right)^2 + \left(\Gamma_{img} - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$$

centre $\left(1, \frac{1}{x}\right)$, radius $\left(\frac{1}{x}\right)$.

Grey Matter Alert!

Full revolution on the smith chart is equivalent to half of the wavelength (λ/2 distance).

x	Centre	Radius
0	(1, ∞)	∞
→ 1	(1, 1)	0
–1	(1, –1)	–1
∞	(1, 0)	0
2	(1, 0.5)	0.5
–2	(1, –0.5)	–0.5

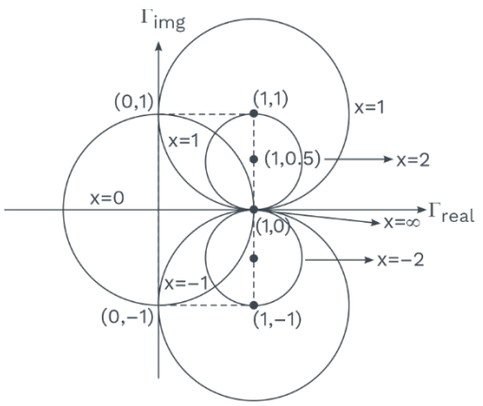


Fig. 6.15 Typical x-Circles for r = 0, ±0.5, ±1, ±2, ±∞.

Smith chart:

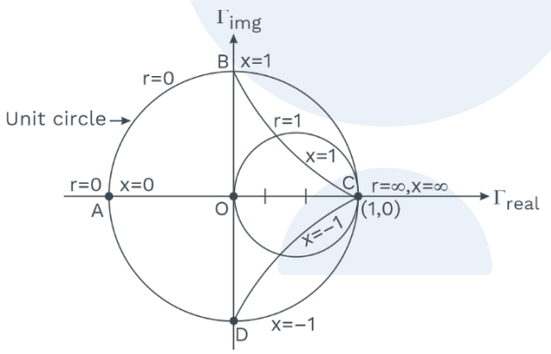


Fig. 6.16 Illustration of the r, x on the Smith Chart

For point A:

$Z_n = r + jx = 0$

$\frac{Z_L}{Z_0} = 0$

$Z_L = 0$ (short circuited)

Point A is denoted short circuited load.

For point B:

$r = 0, x = 1$

$Z_n = 0 + j1$

$Z_L = jZ_0$ (purely inductive)

Point A is denoted purely inductive load.

Grey Matter Alert!

Anticlockwise movement on the smith chart represents a movement towards a 'load'.

For Point C:

$r = \infty, x = \infty$

$Z_n = r + jx = \infty$

$\frac{Z_L}{Z_0} = \infty \Rightarrow Z_L = \infty$ (Open circuited)

Point B is denoted open-circuited load.

For Point D:

$r = 0, x = -1$

$Z_n = -j$

$\frac{Z_L}{Z_0} = -j$

$Z_L = -jZ_0$ (purely capacitive)

Point D is denoted purely capacitive load.

For point O:

$r = 1, x = 0$

$Z_n = 1$

$\frac{Z_L}{Z_0} = 1$

$Z_L = Z_0$ (Matched condition)

Point O is denote matched condition.

Properties of Smith chart:

1) The upper half of the Smith chart represents inductive load.

2) The lower half of the Smith chart represents capacitive load.

3) The clockwise movement on the Smith chart represents a movement towards the source.

4) Anticlockwise movement on the Smith chart represents a movement towards the load.

Calculation of parameter using smith chart:

1) Reflection coefficient:

Let us,

$Z_L = 200 + j100\ \Omega$

$Z_0 = 100\ \Omega$

$$Z_n = \frac{Z_L}{Z_0} = \frac{200 + j100}{100} = 2 + j$$

$r = 2, x = 1$

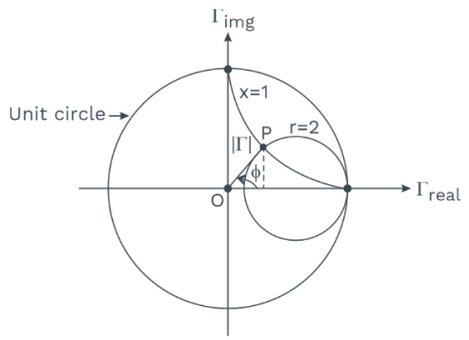


Fig. 6.17 Determining Reflection Coefficient Using the Smith Chart

2) VSWR:

$$S = \frac{1+|\Gamma|}{1-|\Gamma|}$$

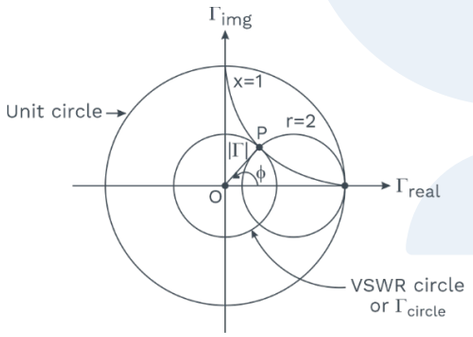


Fig. 6.18 Determining Reflection VSWR Using the Smith Chart

3) The impedance at any distance l' from the load:

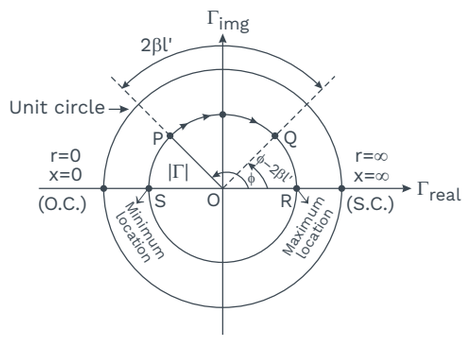


Fig. 6.19 Determining Impedance at Any distance l' from the Load Using the Smith Chart

Previous Years' Question

The impedance Z = jX, for all X in the range (–∞, ∞) map to the Smith chart as:

a) A circle of radius 1 with centre at (0,0).

b) A line passing through the centre of the chart.

c) A circle of radius 0.5 with centre at (0.5, 0).

d) A point at the centre of the chart.

At point Q,

$$\Gamma(l') = \Gamma_{load}\, e^{j(\phi - 2\beta l')}$$

$$l' \uparrow \Rightarrow \theta \downarrow = \left(\phi - 2\beta l' \uparrow\right)$$

$$\therefore \quad \Gamma(l') = \frac{Z(l') - Z_0}{Z(l') + Z_0}$$

4) Location of maxima and minima:

$$2\beta l' = 2\pi$$

$$2 \times \frac{2\pi}{\lambda} \times l' = 2\pi$$

$$l' = \frac{\lambda}{2}$$

where l′ is the distance from the load.
360° rotation on the Smith chart is equivalent to $\frac{\lambda}{2}$ the distance from the load.

At point R in the above figure is:

$$\phi - 2\beta l'_{max} = 0$$

$$\phi = 2\beta l'_{max}$$

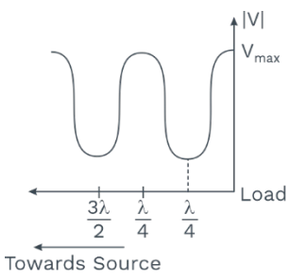
$$l'_{max} = \frac{\phi}{2\beta} \rightarrow \text{By Smith chart}$$

$$l'_{min} = \frac{\phi}{2\beta} + \frac{\lambda}{4} \quad \left\{\begin{array}{l}\text{For } 2\pi \rightarrow \frac{\lambda}{2} \\ \pi \rightarrow \frac{\lambda}{4}\end{array}\right\}$$

180° rotation is equivalent to the λ/4.

BEHAVIOUR OF LOAD

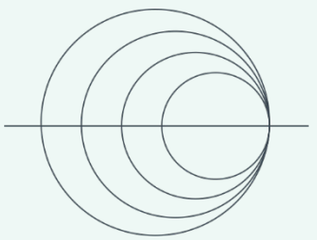
1) If the load impedance is greater than characteristics impedance ($Z_L > Z_0$).



[Resistive Load]

Previous Years' Question

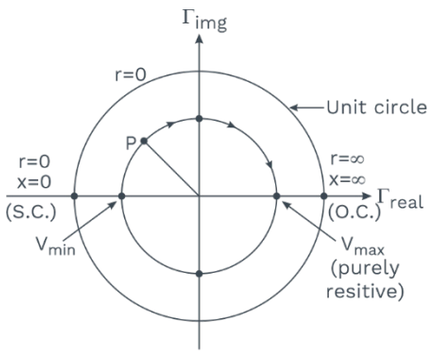
Many circles are drawn in a Smith chart used for transmission line calculations. The circles shown in the figure represent:



a) Unit circles.
b) Constant resistance circles.
c) Constant reactance circles.
d) Constant reflection coefficient circles.

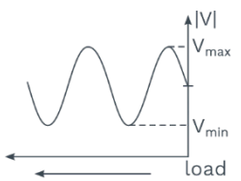
Rack Your Brain

What are the applications of the Smith chart?

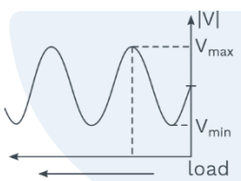


load to source → clockwise direction

2)

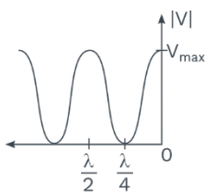


Towards source
[purely inductive load]



Towards source
[purely capacitive load]

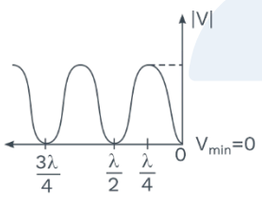
3)



$S = \frac{V_{max}}{V_{min}} = \infty$

$|\Gamma| = \frac{S-1}{S+1} = 1$

V_{max}→Open circuited



$S = \frac{V_{max}}{V_{min}} = \infty$

$|\Gamma| = \frac{S-1}{S+1} = 1$

V_{min}→Short circuited

Note:

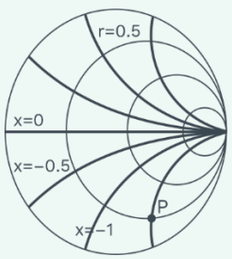
1) Adding an inductance in series is equivalent to clockwise movement in constant resistance circle in impedance Smith chart.

2) Adding a capacitance in series is equivalent to anticlockwise (ACW) movement in constant resistance circle in impedance Smith chart.

3) Adding an inductance in the shunt is equivalent to clockwise movement in a constant admittance circle in the admittance chart.

Previous Years' Question

Consider an impedance Z = R + jX marked with point P in an impedance Smith chart as shown in the below figure. The movement from point P along with a constant resistance circle in the clockwise direction by an angle of 45° is equivalent to:



a) Adding an inductance in series with Z.

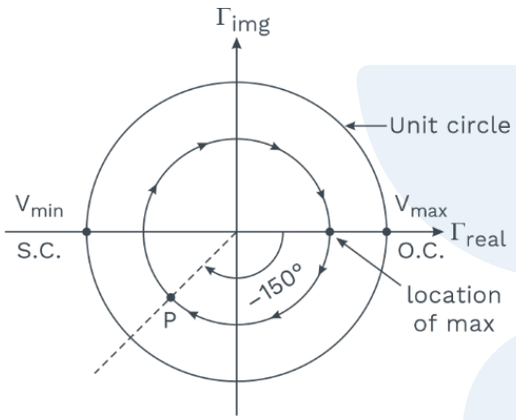
b) Adding a capacitance in series with Z.

c) Adding an inductance in a shunt across Z.

d) Adding a capacitance in a shunt across Z.

4) Adding a capacitance in the shunt is equivalent to anticlockwise (ACW) movement in a constant admittance circle in the admittance chart.

Example 15:
600 m long transmission line has a phase angle –150° of reflection coefficient at the load. If the operating wavelength is 150 m, then the total number of voltage maxima on the line is:
Sol:



Given, $\lambda = 150$ m
We know that,

$\frac{\lambda}{2} \rightarrow 360°$

$\lambda \rightarrow 720°$

$\therefore \quad 150\,\text{m} \rightarrow 720°$

$\therefore \quad 600\,\text{m} \rightarrow \frac{720°}{150} \times 600 = 2880°$

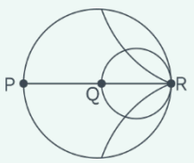
$360° \rightarrow 1\,\text{maxima}$

$2880° \rightarrow \frac{2880}{360} = 8\,\text{maxima}$

$\therefore$ The total number of maxima of the line is 8.

Previous Years' Question

The points P, Q, and R are shown on the Smith chart (normalised impedance chart) in the following figure represents.



a) P: Open circuit, Q: Short circuit, R: Matched load
b) P: Open circuit, Q: Matched load, R: Short circuit
c) P: Short circuit, Q: Matched load, R: Open Circuit
d) P: Short circuit, Q: Open circuit, R: Matched load

TRANSIENT BEHAVIOUR OF TRANSMISSION LINE

Consider a lossless line of length l and characteristic impedance Z_0 as shown in figure 6.20 below. Suppose that the line is driven by a pulse generator of voltage V_g with internal impedance Z_g at z = 0 and terminated with a purely resistive load Z_L. At the instant t = 0 that the switch is closed, the starting current "sees" only Z_g and Z_0, so the initial situation can be described by the equivalent circuit of figure 6.21.

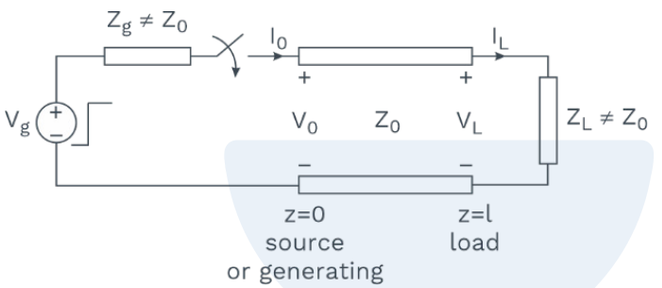


Fig. 6.20 Transient on a Transmission Line Driven by Pulse Generator

At $t = 0^+$

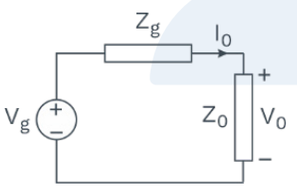


Fig. 6.21 The Equivalent Circuit at z = 0, t = 0^+

By voltage division rule:

$$V_0 = V_g \cdot \frac{Z_0}{Z_0 + Z_g}$$

$$I_0 = \frac{V_g}{Z_0 + Z_g}$$

Transit time: Time is taken by voltage/current wave to travel from source to load and vice versa.

$$t_r = \frac{l}{V_P},\ V_P = \frac{1}{\sqrt{LC}}(\text{for lossless}) = \frac{\omega}{\beta}$$

Voltage bounce diagram: The voltage bounce diagram is shown below.

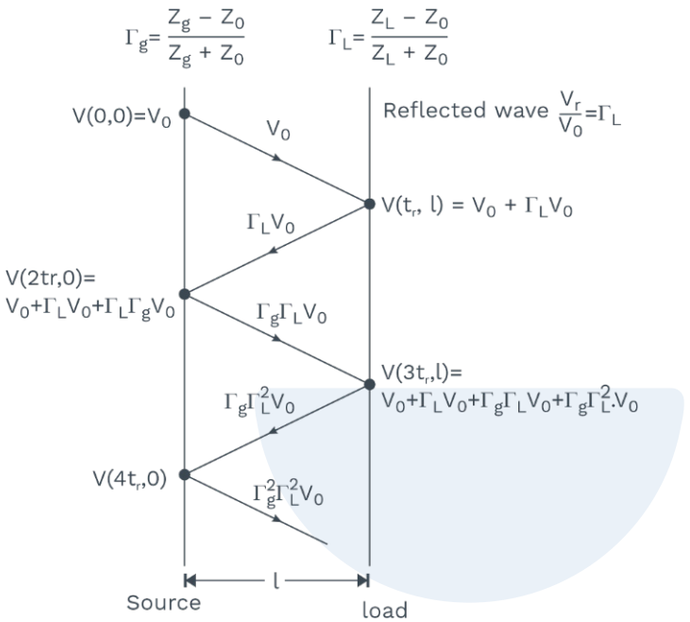


Fig. 6.22 Bounce Diagram for Voltage Wave

Current bounce diagram: Current bounce diagram is shown below.

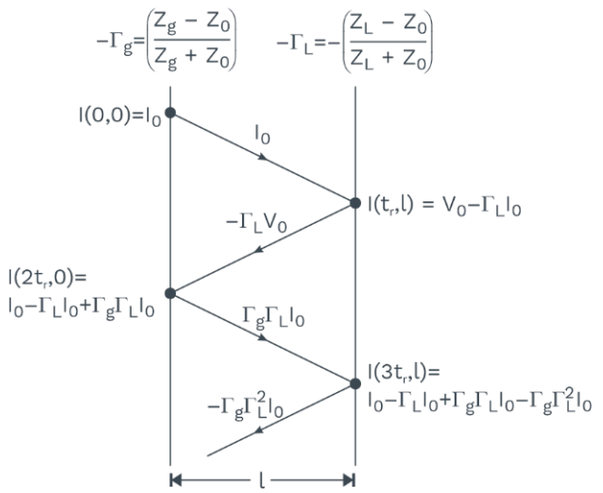


Fig. 6.23 Bounce Diagram for Current Wave

Steady state voltage:

$$V_{SS} = V_0 + \Gamma_L V_0 + \Gamma_g \Gamma_L V_0 + \Gamma_g \Gamma_L^2 V_0 + \Gamma_g^2 \Gamma_L^2 V_0 + \ldots.$$

$$= \left(V_0 + \Gamma_g \Gamma_L V_0 + \Gamma_g^2 \Gamma_L^2 V_0 + \ldots.\right) + \left(\Gamma_L V_0 + \Gamma_g \Gamma_L^2 V_0 + \ldots.\right)$$

$$= \frac{V_0}{1 - \Gamma_g \Gamma_L} + \frac{\Gamma_L V_0}{1 - \Gamma_g \Gamma_L} \quad \left[S_\infty = \frac{a}{1-r}\right]$$

$$V_{SS} = V_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_g \Gamma_L}\right)$$

Steady state current:

$$I_{SS} = I_0 - \Gamma_L I_0 + \Gamma_g \Gamma_L I_0 - \Gamma_g \Gamma_L^2 I_0 + \Gamma_g^2 \Gamma_L^2 I_0 - \Gamma_g^2 \Gamma_L^3 I_0 + \ldots.$$

$$= \left(I_0 + \Gamma_g \Gamma_L I_0 + \Gamma_g^2 \Gamma_L^2 I_0 + \ldots\right) - \left[\Gamma_L I_0 + \Gamma_g \Gamma_L^2 I_0 + \Gamma_g^2 \Gamma_L^3 I_0 + \ldots\right]$$

$$= \frac{I_0}{1 - \Gamma_g \Gamma_L} - \frac{\Gamma_L I_0}{1 - \Gamma_g \Gamma_L}$$

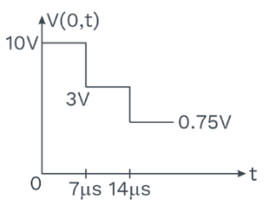
$$I_{SS} = I_0 \left(\frac{1 - \Gamma_L}{1 - \Gamma_g \Gamma_L}\right)$$

Rack Your Brain

What do you understand by return loss?

Example 16:

In response to a step voltage, the waveform shown below was observed at the sending end of a shorted line with $Z_0 = 60\Omega$ and $\in_r = 4$.



Find the:

i) The generator voltage is _____ V.

ii) The generator resistance is _____Ω.

iii) The length of the transmission line is ____ m.

Sol:

Given,

$Z_L = 0$, $Z_0 = 60\Omega$, $\in_r = 4$

$2t_r = 7\ \mu s$

$$t_r = 3.5\ \mu s$$

$$t_r = 3.5\,\mu s = \frac{l}{V_P}$$

$$l = V_P \times 3.5 \times 10^{-6}$$

$$l = \frac{c}{\sqrt{\epsilon_r}} \times 3.5 \times 10^{-6}$$

$$= \frac{3 \times 10^8 \times 3.5 \times 10^{-6}}{\sqrt{4}} = 525\,m$$

$$V_0 = 10V,$$

$$V_0 = \frac{V_g Z_0}{Z_g + Z_0}$$

$$V_g = \frac{V_0 (Z_g + Z_0)}{Z_0}$$

$$V(2t_r, 0) = 3 = V_0 + \Gamma_L V_0 + V_0 \Gamma_g \Gamma_L$$

$$3 = V_0 [1 + \Gamma_L + \Gamma_g \Gamma_L]$$

$$\left\{\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - 60}{60} = -1\right\}$$

$$\frac{3}{10} = 1 - 1 - 1 \times \Gamma_g$$

$$\Gamma_g = -\frac{3}{10}$$

$\therefore$ We know that,

$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0}$$

$$-\frac{3}{10} = \frac{Z_g - 60}{Z_g + 60}$$

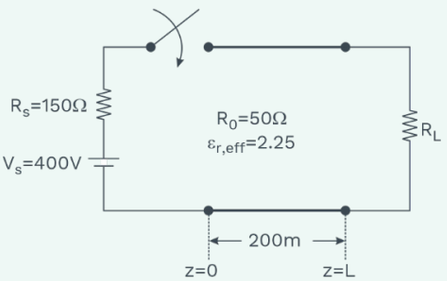
$$-3Z_g - 180 = 10\ Z_g - 600$$

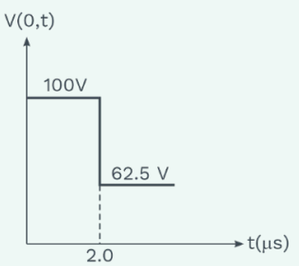
$$13Z_g = 420 \Rightarrow Z_g = 32.3\Omega$$

$$\therefore \quad V_g = \frac{10(60 + 32.3)}{60} = 15.33V$$

Previous Years' Question

A 200 m long transmission line having parameters shown in the figure is terminated with a load R_L. The line is connected to a 400 V source having source resistance R_S through a switch which is closed at t = 0. The transient response of the circuit at the input of the line (z = 0) is also drawn in the figure. The value of R_L (in Ω) is _____.





S-PARAMETER

If the frequency is in the microwave range (1 GHz – 100 GHz) then h, Y, Z parameters cannot be measured for the following reasons.

1) Equipment is not available to measure total voltage and total current at the port of the network.

2) Short circuits and open circuits are difficult to achieve over a broadband frequency.

3) An active device such as a power transistor and tunnel diode will not have stability for open circuit and short circuit conditions.

Due to the above reason, we use the scattering parameter, which is known as S-parameter.

Advantage of S-parameter:

1) S-parameter can be directly measured using a vector network analyser (VNA).

2) For measuring the S-parameter, the ports are neither open nor short-circuited.

3) S-parameter can be obtained for all active, passive, lumped or distributed networks.

The scattering parameter is defined as the ratio of normalised reflected voltage wave at the ith port to the normalised incident voltage wave at the jth port.

$$S_{ij} = \frac{\text{normalised reflected voltage wave at } i^{th} \text{ port}}{\text{normalised incident voltage wave at } j^{th} \text{ port}}$$

$$= \frac{b_i}{a_j}$$

$$b_i = \frac{V_i^-}{\sqrt{Z_{0i}}}, \qquad a_j = \frac{V_j^+}{\sqrt{Z_{0j}}}$$

Definitions

S-Parameter:
The scattering parameter is defined as the ratio of normalised reflected voltage wave at the ith port to the normalised incident voltage wave at the jth port.

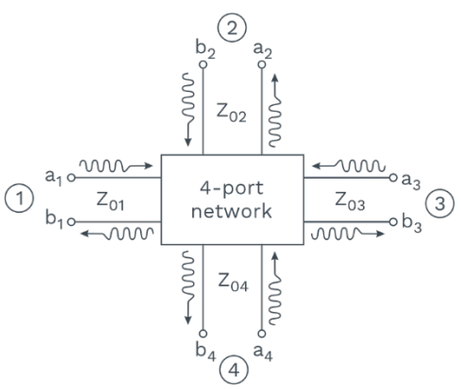


Fig. 6.24 Four-Port Network

$$S_{11} = \left.\frac{b_1}{a_1}\right|_{a_2, a_3, a_4 = 0}$$

$$S_{12} = \left.\frac{b_1}{a_2}\right|_{a_1, a_3, a_4 = 0}$$

$$S_{13} = \left.\frac{b_1}{a_3}\right|_{a_1, a_2, a_4 = 0}$$

$$S_{14} = \left.\frac{b_1}{a_4}\right|_{a_1, a_2, a_3 = 0}$$

$$b_1 = S_{11}a_1 + S_{12}a_2 + S_{13}a_3 + S_{14}a_4$$

Similarly,

$$b_2 = S_{21}a_1 + S_{22}a_2 + S_{23}a_3 + S_{24}a_4$$

$$b_3 = S_{31}a_1 + S_{32}a_2 + S_{33}b_3 + S_{34}a_4$$

$$b_4 = S_{41}a_1 + S_{42}a_2 + S_{43}b_3 + S_{44}a_4$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

$$[b] = [S][a]$$

↳ scattering matrix

$$S_{11} = \frac{\text{Reflected voltage wave at port 1}}{\text{Incident voltage wave at port 1}} = \frac{b_1}{a_1}$$

$$b_1 = \frac{V_0^-}{\sqrt{Z_{01}}}, \qquad a_1 = \frac{V_0^+}{\sqrt{Z_{01}}}$$

$$\frac{b_1}{a_1} = \frac{V_0^-}{V_0^+} = \Gamma$$

∴ Diagonal elements S_{11}, S_{22}, S_{33}, and S_{44} are reflection coefficients at ports 1, 2, 3, and 4, respectively.

For two-port network:



Fig. 6.25 Two-Port Network

$$S_{11} = \frac{b_1}{a_1}, S_{12} = \frac{b_1}{a_2}, S_{21} = \frac{b_2}{a_1}, S_{22} = \frac{b_2}{a_2}$$

$$b_1 = S_{11}a_1 + S_{12}a_2$$
$$b_2 = S_{21}a_1 + S_{22}a_2$$

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

$$b_1 = \frac{V_1^-}{\sqrt{Z_{01}}}, \quad b_2 = \frac{V_2^-}{\sqrt{Z_{02}}}$$

$$a_1 = \frac{V_1^+}{\sqrt{Z_{01}}}, \quad a_2 = \frac{V_2^+}{\sqrt{Z_{02}}}$$

$$S_{11} = \frac{b_1}{a_1} = \frac{V_1^-}{\sqrt{Z_{01}}} \times \frac{\sqrt{Z_{01}}}{V_1^+} = \frac{V_1^-}{V_1^+}$$

$$S_{12} = \frac{b_1}{a_2} = \frac{V_1^-}{\sqrt{Z_{01}}} \times \frac{\sqrt{Z_{02}}}{V_2^+} = \frac{V_1^-}{V_2^+} \quad \{\text{if } Z_{01} = Z_{02}\}$$

$$S_{21} = \frac{b_2}{a_1} = \frac{V_2^-}{\sqrt{Z_{02}}} \times \frac{\sqrt{Z_{01}}}{V_1^+} = \frac{V_2^-}{V_1^+} \quad \{\text{if } Z_{01} = Z_{02}\}$$

$$S_{22} = \frac{b_2}{a_2} = \frac{V_2^-}{\sqrt{Z_{02}}} \times \frac{\sqrt{Z_{02}}}{V_2^+} = \frac{V_2^-}{V_2^+}$$

$$\left.\begin{aligned} V_1^- &= S_{11}V_1 + S_{12}V_2^+ \\ V_2^- &= S_{21}V_1 + S_{22}V_2^+ \end{aligned}\right\} \quad \text{if } Z_{01} = Z_{02} = Z_0$$

Rack Your Brain

What is group velocity?

Properties of S-matrix:

1) S-matrix is a square matrix of order N × N, where, N = number of ports.

2) S-matrix is a symmetric matrix, $S_{ij} = S_{ji}$.

3) S-matrix is a reciprocal, $[S] = [S]^T$.

For a two-port network,

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, [S]^T = \begin{bmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{bmatrix}$$

Hence, $S_{12} = S_{21}$ (for reciprocal)

4) S-matrix is a unitary matrix, [S][S]* = I.

5) The sum of the product of any row (or column) multiplied by the complex conjugate of the corresponding element of any other row (or column) is always equal to zero.

$$\therefore \quad \sum_{i=1}^{N} S_{ij} S_{ik}^{*} = 0, \qquad \text{Where, } j \neq k$$

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad [S]^{*} = \begin{bmatrix} S_{11}^{*} & S_{12}^{*} \\ S_{21}^{*} & S_{22}^{*} \end{bmatrix}$$

1st column → consider 1st column and 2nd column

$$S_{11}S_{12}^{*} + S_{21}S_{22}^{*} = 0$$

Consider 1st row and 2nd row:

$$S_{11}S_{21}^{*} + S_{12}S_{22}^{*} = 0$$

$$\sum_{i=1}^{N} S_{ij} S_{ik}^{*} = 1 \qquad \text{If } j = k$$

6) If any of the reference ports are moved away from the junction by electrical path $\beta_k l_k$ (k^{th} port), then each of coefficient S_{ij} involving K^{th} port multiplied by $e^{-j\beta k l k}$.



$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{22} & S_{22} \end{bmatrix}$$

$$[S] = \begin{bmatrix} S_{11}e^{-j\beta l} & S_{12}e^{-j\beta l} \\ S_{21}e^{-j\beta l} & S_{22} \end{bmatrix}$$

7) S-matrix is said to be lossless

$[S]^{T}[S]^{*} = I$

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}, \quad [S]^{T} = \begin{bmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{bmatrix}$$

$$[S]^{*} = \begin{bmatrix} S_{11}^{*} & S_{12}^{*} \\ S_{21}^{*} & S_{22}^{*} \end{bmatrix}$$

Previous Years' Question

A transmission line with a characteristic impedance of 100Ω is used to match a 50Ω section to a 200Ω section. If the matching is to be done both at 429 MHz and 1 GHz, the length of the transmission line can be approximately.

a) 82.5 cm
b) 1.05 m
c) 1.58 m
d) 1.75 m

$$\begin{bmatrix} S_{11} & S_{21} \\ S_{12} & S_{22} \end{bmatrix}\begin{bmatrix} S_{11}^* & S_{12}^* \\ S_{21}^* & S_{22}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S_{11}S_{11}^* + S_{21}S_{21}^* = 1$$

$$|S_{11}|^2 + |S_{21}|^2 = 1$$

$$S_{12}S_{12}^* + S_{22}S_{22}^* = 1$$

$$|S_{12}|^2 + |S_{22}|^2 = 1$$

Example 17:

If the scattering matrix [S] of a two-port network is $[S] = \begin{bmatrix} 0.1\angle 0° & 0.8\angle 90° \\ 0.8\angle 90° & 0.1\angle 90° \end{bmatrix}$. Whether the network is lossy?

Sol:

Given, $[S] = \begin{bmatrix} 0.1\angle 0° & 0.8\angle 90° \\ 0.8\angle 90° & 0.1\angle 90° \end{bmatrix}$

$S_{12} = S_{21} = 0.8\angle 90°$ [Reciprocal]

Lossless condition is-

$|S_{11}|^2 + |S_{21}|^2 = 1$

$|0.1|^2 + |0.8|^2 = 0.01 + 0.64 = 0.65 \neq 1$

Hence, network is lossy and reciprocal.

Rack Your Brain

What are the properties of S-parameters?

Example 18:

A 4-port network has the scattering matrix as shown below:

$$[S] = \begin{bmatrix} 0.2\angle 90° & 0.8\angle -45° & 0.3\angle -45° & 0 \\ 0.8\angle -45° & 0 & 0 & 0.5\angle 45° \\ 0.3\angle -45° & 0 & 0 & 0.6\angle -45° \\ 0 & 0.5\angle 45° & 0.5\angle -45° & 0 \end{bmatrix}$$

i) Is this network reciprocal?

ii) Is this network lossless?

iii) What is return loss [RL] at port 1 when all other ports are terminated with a matched load?

Sol:

i) Given,

$$[S] = \begin{bmatrix} 0.2\angle 90° & 0.8\angle -45° & 0.3\angle -45° & 0 \\ 0.8\angle -45° & 0 & 0 & 0.5\angle 45° \\ 0.3\angle -45° & 0 & 0 & 0.6\angle -45° \\ 0 & 0.5\angle 45° & 0.5\angle -45° & 0 \end{bmatrix}$$

$$[S]^T = \begin{bmatrix} 0.2\angle 90° & 0.8\angle -45° & 0.3\angle -45° & 0 \\ 0.8\angle -45° & 0 & 0 & 0.5\angle 45° \\ 0.3\angle -45° & 0 & 0 & 0.5\angle -45° \\ 0 & 0.5\angle 45° & 0.6\angle -45° & 0 \end{bmatrix}$$

Here, $[S] \neq [S]^T$, not reciprocal.

i) For lossless

$$|S_{11}|^2 + |S_{21}|^2 + |S_{31}|^2 + |S_{41}|^2 = 1$$

$$|0.2|^2 + |0.8|^2 + |0.3|^2 + |0|^2$$

$0.04 + 0.64 + 0.09 + 0 = 0.77 \neq 1$

Hence, the network is lossy.

ii) $b_1 = S_{11}a_1 + S_{12}a_2 + S_{13}a_3 + S_{14}a_4$

when all other ports other than port 1 are matched terminated.

$a_1 \neq 0$, $a_2 = a_3 = a_4 = 0$

$S_{11} = \dfrac{b_1}{a_1}$ = reflection coefficient

$S_{11} = 0.2\angle 90°$

Return does $= -20 \log |\Gamma|$

$= -20 \log |0.2|$

$= 13.97$ dB

S-Parameter for series element:

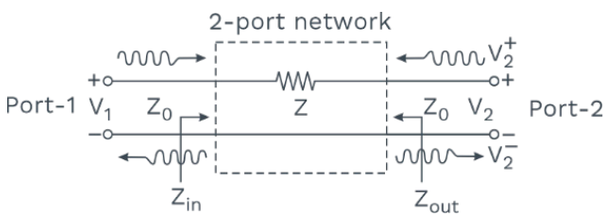


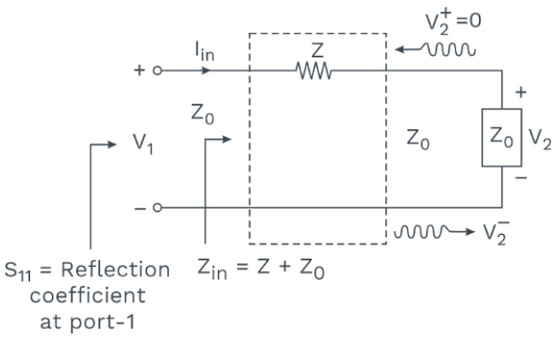
Fig. 6.26 S-Parameter for Two-Port Series Element

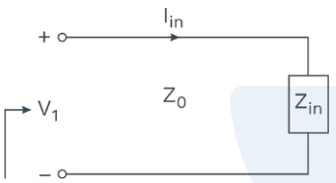
Since, the characteristic impedance of both ports are the same.

$$V_1^- = S_{11}V_1^+ + S_{12}V_2^+$$

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^+$$

$$S_{11} = \left.\frac{V_1^-}{V_1^+}\right|_{V_2^+=0}, \quad S_{21} = \left.\frac{V_2^-}{V_1^+}\right|_{V_2^+=0}$$





$$S_{11} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{Z + Z_0 - Z_0}{Z + Z_0 + Z_0} = \frac{Z}{Z + 2Z_0}$$

$$V_2 = I_{in} Z_0$$

$$V_2^+ + V_2^- = I_{in} Z_0$$

$$0 + V_2^- = I_{in} Z_0$$

or $$V_2^- = I_{in} Z_0 \quad \text{...(i)}$$

From above figure:

$$I_{in} = \frac{V_1}{Z_{in}} \quad \text{...(ii)}$$

or $V_1 = I_{in} Z_{in}$

$$V_1^+ + V_1^- = I_{in}(Z + Z_0)$$

$$\frac{V_1^-}{V_1^+} + 1 = \frac{I_{in}}{V_1^+}(Z + Z_0)$$

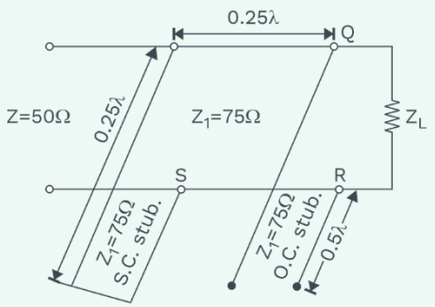
$$S_{11} + 1 = \frac{V_2^-}{Z_0}\left(\frac{Z + Z_0}{V_1^+}\right)$$

$$S_{11} + 1 = S_{21}\left(\frac{Z + Z_0}{Z_0}\right)$$

$$S_{21} = (1 + S_{11})\left(\frac{Z_0}{Z + Z_0}\right)$$

Previous Years' Question

The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50\Omega$ with a load impedance Z_L. A quarter-wave line having a characteristic impedance $Z_1 = 75\Omega$ is connected to Z_L. Two stubs having a characteristic impedance of 75Ω each are connected to this quarter-wave line. One is a short-circuited (S.C.) stub of length 0.25λ connected across PS and the other one is an open-circuited (O.C.) stub of length 0.5λ connected across QR.



The impedance matching is achieved when the real part of Z_L is:

a) 112.5Ω **b)** 75.0Ω
c) 50.0Ω **d)** 33.3Ω

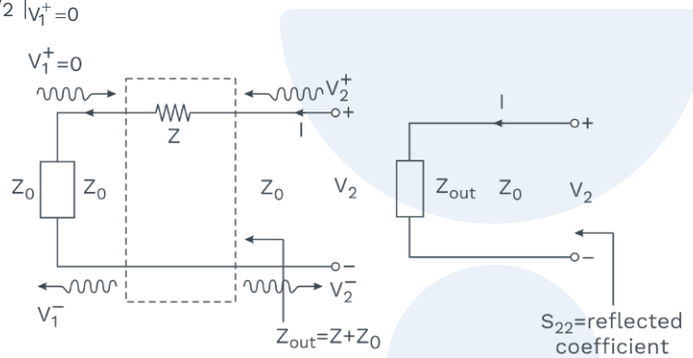
$$= \left(1 + \frac{Z}{Z + 2Z_0}\right)\left(\frac{Z_0}{Z + Z_0}\right)$$

$$= \frac{2(Z + Z_0)Z_0}{(Z + 2Z_0)(Z + Z_0)}$$

$$= \frac{2Z_0}{Z + 2Z_0}$$

$$S_{12} = \left.\frac{V_1^-}{V_2^+}\right|_{V_1^+ = 0}$$

$$S_{22} = \left.\frac{V_2^-}{V_2^+}\right|_{V_1^+ = 0}$$



$$S_{22} = \frac{Z_{out} - Z_0}{Z_{out} + Z_0}$$

$$S_{22} = \frac{Z + Z_0 - Z_0}{Z + Z_0 + Z_0} = \frac{Z}{Z + 2Z_0} = \frac{V_2^-}{V_2^+}$$

$$V_2 = I\, Z_{out}$$

$$V_2^+ + V_2^- = I Z_{out}$$

$$\frac{V_2^-}{V_2^+} + 1 = \frac{I}{V_2^+} Z_{out}$$

$$S_{22} + 1 = \frac{I Z_{out}}{V_2^+}$$

$$1 + S_{22} = \frac{V_1^-}{Z_0}\frac{(Z + Z_0)}{V_2^+} \qquad \left\{\begin{array}{l} V_1 = IZ_0 \\ V_1^+ + V_1^- = IZ_0 \\ 0 + V_1^- = IZ_0 \\ V_1^- = IZ_0 \\ I = \dfrac{V_1^-}{Z_0} \end{array}\right\}$$

$$\left(1 + \frac{Z}{Z + 2Z_0}\right)\frac{Z_0}{(Z + Z_0)} = \frac{V_1^-}{V_2^+}$$

$$S_{12} = \frac{2(Z + Z_0)Z_0}{(Z + 2Z_0)(Z + Z_0)} = \frac{2Z_0}{Z + 2Z_0}$$

S-Parameter for the series element is:

$$[S] = \begin{bmatrix} \dfrac{Z}{Z + 2Z_0} & \dfrac{2Z_0}{Z + 2Z_0} \\ \dfrac{2Z_0}{Z + 2Z_0} & \dfrac{Z}{Z + 2Z_0} \end{bmatrix}$$

S-Parameter for shunt element:

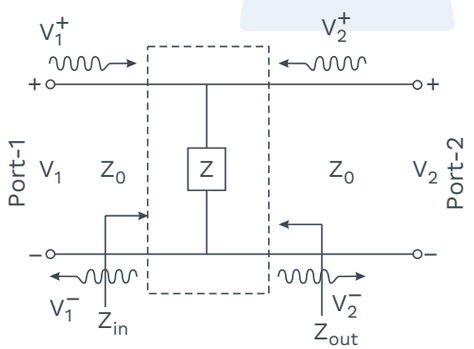
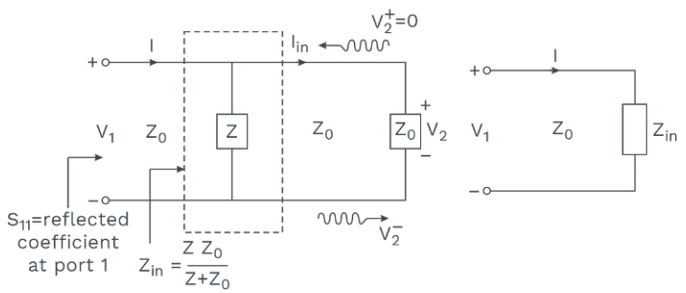


Fig. 6.27 S-Parameter for Two-Port Shunt Element

$$V_1^- = S_{11}V_1^+ + S_{12}V_2^+$$

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^+$$

$$S_{11} = \left.\frac{V_1^-}{V_1^+}\right|_{V_2^+ = 0}, \quad S_{21} = \left.\frac{V_2^-}{V_1^+}\right|_{V_2^+ = 0}$$



$$S_{11}=\frac{Z_{in}-Z_0}{Z_{in}+Z_0}=\frac{\frac{ZZ_0}{Z+Z_0}-Z_0}{\frac{ZZ_0}{Z+Z_0}+Z_0}$$

$$=\frac{ZZ_0-ZZ_0-Z_0^2}{ZZ_0+ZZ_0+Z_0^2}$$

$$=\frac{-Z_0^2}{2ZZ_0+Z_0^2}=\frac{-Z_0}{2Z+Z_0}$$

$$V_1=I\,Z_{in}$$

$$V_1^+ + V_1^- = I Z_{in}$$

$$\frac{V_1^+}{V_1^+}+\frac{V_1^-}{V_1^+}=\frac{I}{V_1^+}Z_{in}$$

$$1+S_{11}=\frac{I}{V_1^+}Z_{in} \qquad ...(i)$$

$$V_2 = I_{in}\,Z_0$$

$$V_2^+ + V_2^- = I_{in}Z_0$$

$$0 + V_2^- = I_{in}Z_0$$

$$V_2^- = I_{in}Z_0$$

$$V_2^- = I\left(\frac{Z}{Z+Z_0}\right)Z_0$$

$$I=\frac{V_2^-}{ZZ_0}(Z+Z_0) \qquad ...(ii)$$

From equation i) and ii), we get:

$$1+S_{11}=\frac{V_2^-(Z+Z_0)}{(ZZ_0)V_1^+}Z_{in}$$

Previous Years' Question

The magnitudes of the open-circuit and short-circuit input impedance of a transmission line are 100Ω and 25Ω, respectively.
The characteristic impedance of the line is:

a) 25 Ω **b)** 50 Ω
c) 75 Ω **d)** 100 Ω

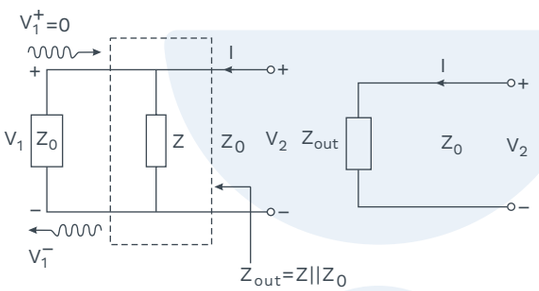
$$1 + S_{11} = S_{21}\left(\frac{Z + Z_0}{ZZ_0}\right)\left(\frac{ZZ_0}{Z + Z_0}\right)$$

$$1 + S_{11} = S_{21}$$

$$S_{21} = 1 + \left(\frac{-Z_0}{2Z + Z_0}\right)$$

$$S_{21} = \frac{2Z}{2Z + Z_0}$$

$$S_{12} = \frac{V_1^-}{V_2^+}\bigg|_{V_1^+ = 0}, \quad S_{22} = \frac{V_2^-}{V_2^+}\bigg|_{V_1^+ = 0}$$



$$S_{22} = \frac{Z_{out} - Z_0}{Z_{out} + Z_0}$$

$$= \frac{\frac{ZZ_0}{Z + Z_0} - Z_0}{\frac{ZZ_0}{Z + Z_0} + Z_0}$$

$$= \frac{ZZ_0 - ZZ_0 - Z_0^2}{ZZ_0 + ZZ_0 + Z_0^2}$$

$$= \frac{-Z_0^2}{2ZZ_0 + Z_0^2}$$

$$S_{22} = \frac{-Z_0}{2Z + Z_0}$$

$$V_2 = I\, Z_{out}$$

$$V_2^+ + V_2^- = I\, Z_{out}$$

$$1 + S_{22} = \frac{I}{V_2^+} Z_{out}$$

Grey Matter Alert!

Balun = balanced to unbalanced load transformer
A balun is an electrical device that converts a balanced signal into an unbalanced signal.
Balun lines are used to connect lines of differing impedance.

$$1 + S_{22} = \frac{V_1^-}{V_2^+} = S_{12}$$

$$S_{12} = 1 - \frac{Z_0}{2Z + Z_0}$$

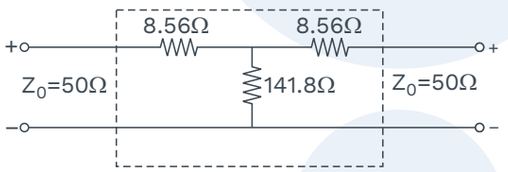
$$S_{12} = \frac{2Z}{2Z + Z_0}$$

$$[S] = \begin{bmatrix} \frac{-Z_0}{2Z + Z_0} & \frac{2Z}{2Z + Z_0} \\ \frac{2Z}{2Z + Z_0} & \frac{-Z_0}{2Z + Z_0} \end{bmatrix}$$

This is S-matrix for the shunt element.

Example 19:

Find the S-matrix for the following two-port network.

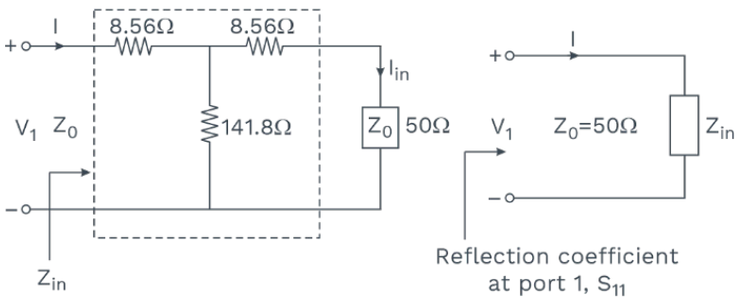


Sol:

$$V_1^- = S_{11}V_1^+ + S_{12}V_2^+$$

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^+$$

$$S_{11} = \left.\frac{V_1^-}{V_1^+}\right|_{V_2^+ = 0}, \quad S_{21} = \left.\frac{V_2^-}{V_1^+}\right|_{V_2^+ = 0}$$



$$Z_{in} = (58.56 \parallel 141.8) + 8.56 = 50\Omega$$

$$S_{11} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{50 - 50}{50 + 50} = 0$$

$$V_1 = I\, Z_{in}$$

$$V_1^+ + V_1^- = I Z_{in}$$

$$1 + S_{11} = \frac{I}{V_1^+} Z_{in} \qquad \ldots(i)$$

$$V_2 = I_{in} Z_0 \qquad \Rightarrow V_2^+ + V_2^- = I_{in} Z_0$$

$$V_2^- = \frac{I \times 141.8}{141.8 + 58.56} Z_0 \qquad \{V_2^+ = 0\}$$

$$I = \frac{V_2^- (141.8 + 58.56)}{141.8 \times 50}$$

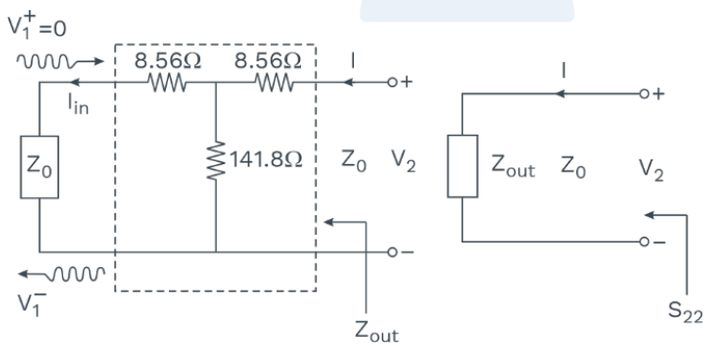
$$I = 0.0282\, V_2^- \qquad \ldots(ii)$$

From equation i) and ii), we get:

$$1 + 0 = \frac{0.0282 V_2^-}{V_1^+} \times 50$$

$$1 = 1.413\ S_{21}$$

$$S_{21} = 0.707 = \frac{1}{\sqrt{2}}$$



$$S_{22} = \frac{Z_{out} - Z_0}{Z_{out} + Z_0}$$

$$Z_{out} = (58.56 \,||\, 141.8) + 8.56 = 50\Omega$$

$$\therefore \quad S_{22} = \frac{50 - 50}{50 + 50} = 0$$

$$V_2 = I\, Z_{out}$$

$$V_2^- + V_2^+ = I Z_{out}$$

$$1 + S_{22} = \frac{I}{V_2^+} 50 \qquad \text{...(iii)}$$

$$V_1 = I_{in}\, Z_0$$

$$V_1^+ + V_1^- = I_{in} \times Z_0$$

$$0 + V_1^- = I\left(\frac{141.8}{141.8 + 58.56}\right) \times 50$$

$$I = V_1^-\left(\frac{141.8 + 58.56}{141.8 \times 50}\right) \qquad \text{...(iv)}$$

From equation iii) and iv), we get:

$$1 + 0 = \frac{V_1^-}{V_2^+}\left(\frac{141.8 + 58.56}{141.8 \times 50}\right) 50$$

$$1 = S_{12}\ 1.413$$

$$S_{12} = 0.707 = \frac{1}{\sqrt{2}}$$

$$[S] = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Relationship between S-parameter and ABCD parameter:

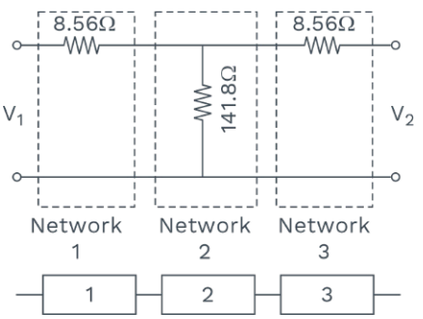
$$S_{11} = \frac{A + \frac{B}{Z_0} - CZ_0 - D}{A + \frac{B}{Z_0} + CZ_0 + D}$$

$$S_{12} = \frac{2(AD - BC)}{A + \frac{B}{Z_0} + CZ_0 + D}$$

$$S_{21} = \frac{2}{A + \frac{B}{Z_0} + CZ_0 + D}$$

$$S_{22} = \frac{-A + \frac{B}{Z_0} - CZ_0 + D}{A + \frac{B}{Z_0} + CZ_0 + D}$$

Example 20:
Find the S-parameter using T-parameter.

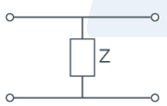


Sol:
For series element:



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

For shunt element:



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z} & 1 \end{bmatrix}$$

$\therefore$ Network 1,2,3 are cascaded.

$$[T] = \begin{bmatrix} 1 & 8.56 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ \frac{1}{141.8} & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 8.56 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 8.56 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 8.56 \\ \frac{1}{141.8} & \frac{8.56}{141.8} + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + \frac{8.56}{141.8} & 8.56 + 8.56\left(\frac{8.56}{141.8} + 1\right) \\ \frac{1}{141.8} & \frac{8.56}{141.8} + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1.06 & 17.63 \\ 7.05 \times 10^{-3} & 1.06 \end{bmatrix}$$

$A = 1.06,\ B = 17.63$

$C = 7.05 \times 10^{-3} = \dfrac{1}{141.8},\ D = 1.06$

$$S_{11} = \frac{A + \dfrac{B}{Z_0} - CZ_0 - D}{A + \dfrac{B}{Z_0} + CZ_0 + D}$$

$$= \frac{1.06 + \dfrac{17.63}{50} - \dfrac{1}{141.8} \times 50 - 1.06}{1.60 + \dfrac{17.63}{50} + \dfrac{1}{141.8} \times 50 + 1.06} = 0$$

$$S_{12} = \frac{2(AD - BC)}{A + \dfrac{B}{Z_0} + CZ_0 + D}$$

$$= \frac{2\left(1.06 \times 1.06 - \dfrac{17.63}{141.8}\right)}{1.06 + \dfrac{17.63}{50} + \dfrac{1}{141.8} + 1.06} = \frac{1}{\sqrt{2}}$$

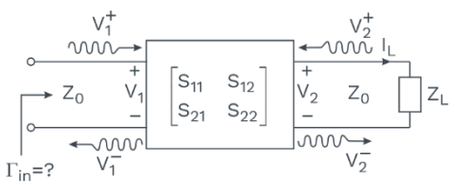
$$S_{21} = \frac{2}{A + \dfrac{B}{Z_0} + CZ_0 + D}$$

$$= \frac{2}{1.06 + \dfrac{17.63}{50} + \dfrac{1}{141.8} + 1.06} = \frac{1}{\sqrt{2}}$$

$$S_{22} = \frac{-A + \dfrac{B}{Z_0} - CZ_0 + D}{A + \dfrac{B}{Z_0} + CZ_0 + D}$$

$$= \frac{-1.06 + \dfrac{17.63}{50} - \dfrac{1}{141.8} \times 50 + 1.06}{1.06 + \dfrac{17.63}{50} + \dfrac{1}{141.5} \times 50 + 1.06} = 0$$

Example 21:
Find the input reflection coefficient.



Sol:

$$V_1^- = S_{11}V_1^+ + S_{22}V_2^+ \quad \text{...(i)}$$

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^+ \quad \text{...(ii)}$$

$$\Gamma_{in} = \frac{V_1^-}{V_1^+}, \qquad \Gamma_{load} = \frac{V_2^+}{V_2^-}$$

$$\therefore \quad V_2^+ = \Gamma_{load}V_2^-$$

From equation ii), we get:

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^-\,\Gamma_{load}$$

$$V_2^-\left[1 - S_{22}\Gamma_{load}\right] = S_{21}V_1^+$$

$$\frac{V_2^+}{\Gamma_{load}}\left[1 - S_{22}\Gamma_{load}\right] = S_{21}V_1^+$$

$$V_2^+ = \frac{S_{21}V_1^+\Gamma_{load}}{\left[1 - S_{22}\Gamma_{load}\right]}$$

From equation i), we get:

$$V_1^- = S_{11}V_1^+ + \frac{S_{12}S_{21}V_1^+\Gamma_{load}}{1 - S_{22}\Gamma_{load}}$$

$$\frac{V_1^-}{V_1^+} = S_{11} + \frac{S_{12}S_{21}\Gamma_{load}}{1 - S_{22}\Gamma_{load}}$$

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_{load}}{1 - S_{22}\Gamma_{load}}$$

Chapter Summary



1) A transmission line is commonly described by its distributed parameters R (in Ω/m), L (in H/m), G (in S/m), and C(in F/m).

2) The distributed parameters are used in an equivalent circuit model to represent a differential length of the line. The transmission-line equations are obtained by applying Kirchhoff's laws and allowing the length of the line to approach zero. The voltage and current waves on the line are:

$$V_s(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$

$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

3) The characteristic impedance Z_0 of a line is given as:

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

And the propagation constant γ (in per meter) is given as:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

The wavelength and wave velocity are:

$$\lambda = \frac{2\pi}{\beta}, \quad v_p = \frac{\omega}{\beta} = \lambda f$$

4) The general case for the lossy transmission line is $R \neq 0 \neq G$ consider. For a lossless line, R = 0, G = 0; for a distortionless line, $\frac{R}{L} = \frac{G}{C}$. Power lines should be lossless and telephone lines are distortionless.

5) The voltage reflection coefficient at the load end is defined as

$$\Gamma_L = \frac{V_0^-}{V_0^+}\left(\frac{Z_L - Z_0}{Z_L + Z_0}\right)$$

Where Z_L = load impedance
And the standing wave ratio is:

$$SWR = \frac{1+|\Gamma_{load}|}{1-|\Gamma_{load}|}$$

6) At any point on the line, the ratio of the phasor voltage to phasor current is the impedance at that point looking towards the load and would be the input impedance to the line if the line were that long. For a lossy line,

$$Z_{in} = Z_0\left[\frac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l}\right]$$

7) Where l is the distance from the load to the point. For a lossless line $\alpha = 0$, tanh γl = tanh βl, for shorted line, $Z_L = 0$; for an open-circuited line, $Z_L = \infty$ and for a matched line, $Z_L = Z_0$.
The Smith chart is a graphical means of obtaining line characteristics such as Γ_L, S and Z_{in}. It is constructed within a circle of unit radius and based on the formula for Γ_L given above. For each r and x, it has two explicit circles (the resistance and reactance circles) and one implicit circle (the constant s-circle). It is conveniently used in determining the location of a stub tuner and its length. It is also used with the slotted line to determine the value of the unknown load impedance.

8) When a dc voltage is suddenly applied at the sending end of a line, a pulse moves forth and back on the line. The transient behaviour is conveniently analyzed using the bounce diagrams.

PRACTICE QUESTIONS

Q1 **Which of the following statement is not correct about the line parameters R, L, G, and C?**

a) **G and C are shunt elements.**

b) **R and L are series elements.**

c) $G = \frac{1}{R}$.

d) **The parameter is not lumped but distributed.**

Q2 **A lossless transmission line of length 50 cm with L = 10 μH/m, C = 40pF/m is operated at 30 MHz. It electrical length is:**

a) **40 π**

b) **108°**

c) **10 λ**

d) **None of these**

Q3 **The characteristics impedance of lossy transmission line does not depend on:**

a) **The operating frequency of the line.**

b) **The conductivity of the conductors.**

c) **The load terminates the line.**

d) **The conductivity of the dielectric separating the conductor.**

Q4 **Which of the following is correct about the lossless line?**

a) $Z_{in} = -jZ_0$ **for shorted line with** $l = \frac{\lambda}{8}$

b) $Z_{in} = j\infty$ **for a shorted with** $l = \frac{\lambda}{4}$

c) $Z_{in} = Z_0$ **for a matched line.**

d) **At a** $\frac{\lambda}{2}$ **distance from load,** $Z_{in} = Z_L$ **and repeats for every half wavelength thereafter.**

Q5 **In airline line, adjacent maxima are found 12.5 cm and 37.5 cm. Find the operating frequency.**

a) 200 MHz **b) 100 GHz**
c) 600 MHz **d) 1.2 GHz**

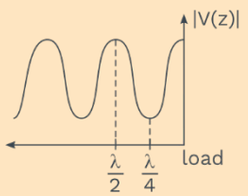
Q6 **Antenna with impedance 40 + j30 Ω is to be matched to a 100 Ω lossless line with a shorted stub. Find the required stub admittance.**

a) j5 mS **b) –3j mS**
c) ±j23 mS **d) ±j10.4 mS**

Q7 **A transmission line is terminated by a load with admittance Y_L = (0.6 + j0.8)/ Z_0. Find the normalised input impedance $\frac{\lambda}{6}$ from the load.**

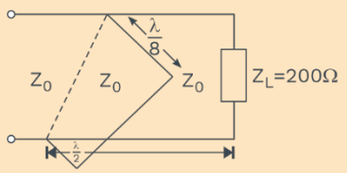
a) 0.2 + j0.5 **b) 3 + j0.45**
c) (0.35 + j0.24) **d) 1 + j**

Q8 **Voltage standing wave pattern in a lossless transmission line with characteristic impedance as 100Ω is shown below. Find the maximum input impedance, if the return loss is 20 dB.**



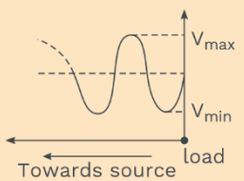
a) 122Ω **b) 84Ω**
c) 100Ω **d) 82Ω**

Q9 **A short-circuited stub is a shunt connected to a transmission line, as shown below. If $Z_0 = 100\Omega$, the admittance Y is seen at the junction of the stub of the transmission line.**



a) (0.005 + j0.01) ℧
b) (0.05 + j0.01) ℧
c) 0.02 ℧
d) (0.02 + j0.01) ℧

Q10 **The voltage standing wave pattern at the lossless transmission line is shown below. Identify the type of load.**

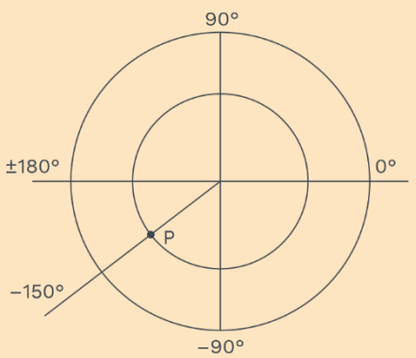


a) Open circuit at load
b) Short circuit at load
c) Capacitive
d) Inductive

Q11 **Two identical pulses each of magnitude 12V and width 2 μs are incident at t = 0 on a lossless transmission line of length 400 m terminated with a load if the two pulses are separated 3 μs, and the phase velocity = 2×10^8 m/s, when does the contribution to $V_L(l, t)$ by the second pulse start overlapping that of the first?**

a) 6 μs
b) 3 μs
c) 2 μs
d) 1 μs

Q12 **A 500-m lossless transmission line is terminated by a load which is located at P on the smith chart as shown below. If λ = 150 m, how many voltage maxima exist on the line?**



a) 5 **b) 4**
c) 6 **d) 7**

Q13 **A transmission line with a characteristic impedance of 50Ω is connected to a 100Ω resistive load. Calculate the voltage reflection at the load.**

a) 0.333 **b) 0.1**
c) 0.25 **d) 0.54**

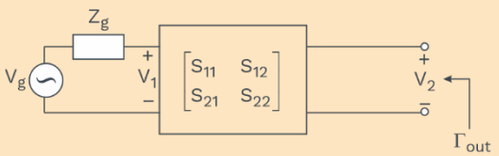
Q14 **A line 10 km long has the following constants.**

$$\alpha = 0.1 \text{ Nepers/km}, \beta = 0.05 \text{ rad/km}, Z_0 = 600\angle 0° \ \Omega$$

Find the received current when 20 mA is sent into one end and receiving end is short-circuited.

a) $13\angle -20°$ mA **b) $5\angle -50°$ mA**
c) $13.66\angle -22.3°$ mA **d) $13\angle -60°$ mA**

Q15 **Find the output reflection coefficient.**



a) $\Gamma_{out} = S_{22} + \dfrac{S_{12}S_{21}\Gamma_g}{1-\Gamma_g S_{11}}$

b) $\Gamma_{out} = S_{11} + \dfrac{S_{12}S_{21}\Gamma_g}{1-\Gamma_g S_{11}}$

c) $\Gamma_{out} = S_{22} + \dfrac{S_{12}S_{21}\Gamma_g}{1+\Gamma_g S_{11}}$

d) None of these

Answer key:

Q1 c)	**Q2 b)**	**Q3 c)**	**Q4 a)**
Q5 c)	**Q6 d)**	**Q7 c)**	**Q8 a)**
Q9 b)	**Q10 c)**	**Q11 a)**	**Q12 d)**
Q13 a)	**Q14 c)**	**Q15 a)**	